

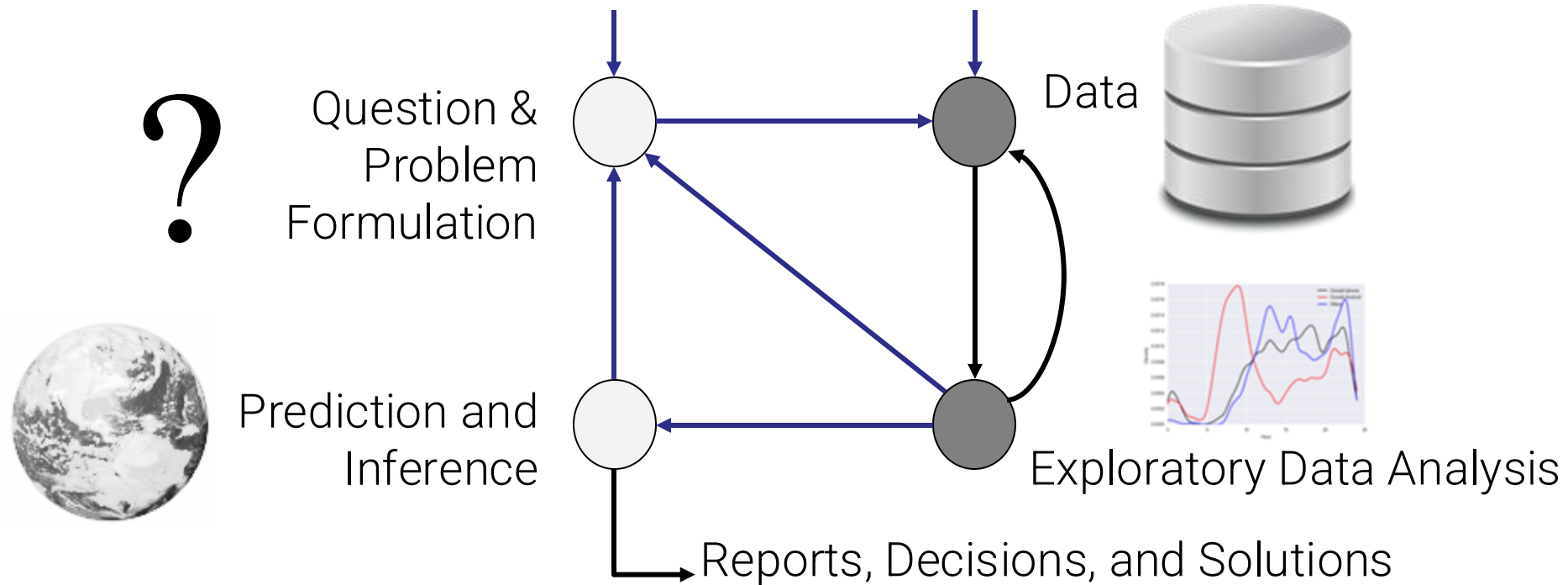
Introduction to Data Science

Visualization



Usman Nazir

Agenda



(today)

Data Wrangling
Intro to EDA



Working with Text Data
(Regular Expressions)

Plots and variables
Seaborn

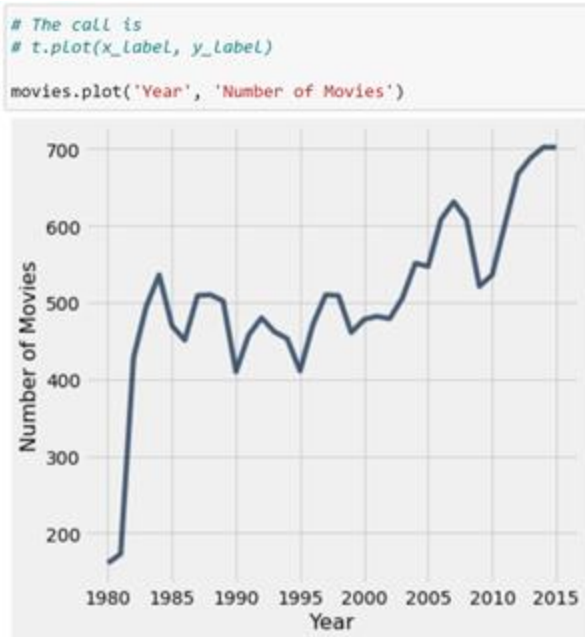
Viz principles
KDE/Transformations

(Part I: Processing Data)

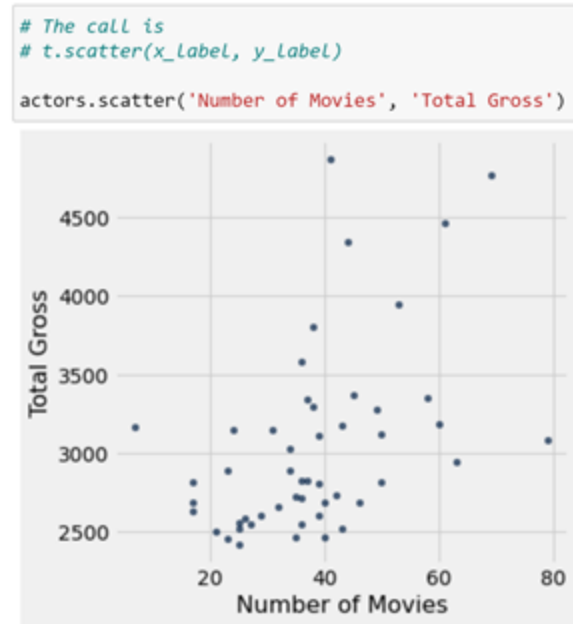
(Part II: Visualizing and Reporting Data)

Visualizations (so far)

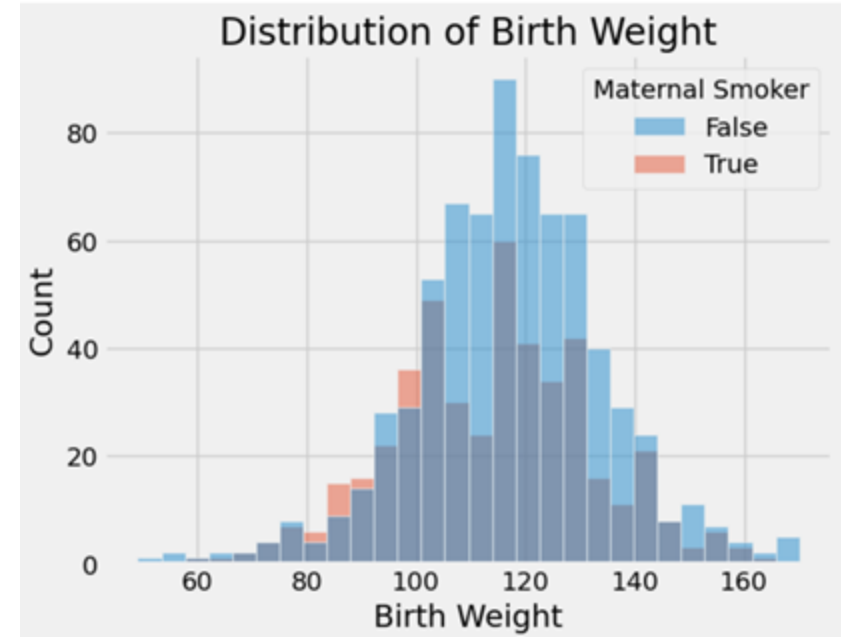
You worked with many types of visualizations



Line plot



Scatter plot



Histogram

What did these achieve?

- Provide a high-level overview of a complex dataset.
- Communicated trends to viewers.

Goals of Data Visualization

Goal 1: To help your own understanding of your data/results.

- Key part of exploratory data analysis.
- Summarize trends visually before in-depth analysis.
- Lightweight, iterative and flexible.

Goal 2: To communicate results or conclusions to others.

- Highly editorial and selective.
- Be thoughtful and careful!
- Fine-tuned to achieve a communications goal.
- Considerations: clarity, accessibility, and necessary context.

What do these goals imply?

Visualizations aren't a matter of making "pretty" pictures.

We need to do a lot of thinking about what stylistic choices communicate ideas most effectively.

Goals of Data Visualization

What do these goals imply?

Visualizations aren't a matter of making "pretty" pictures.

We need to do a lot of thinking about what stylistic choices communicate ideas most effectively.



First half of visualization topics:

Choosing the "right" plot for

- Introducing plots for different variable types
- Generating these plots through code

Second half of visualization topics:

Stylizing plots appropriately

- Smoothing and transforming visual data
- Providing context through labeling and color

Visualizing Distributions

- Goals of visualization
- **Visualizing distributions**

Distributions

A distribution describes...

- The set of values that a variable can possibly take.
- The frequency with which each value occurs.

...for a **single** variable

Example: Distribution of faculty to different departments at Beaconhouse National University.

- The list of departments at BNU.
- The number of faculty in each department.

In other words: How is the variable distributed across all of its possible values?

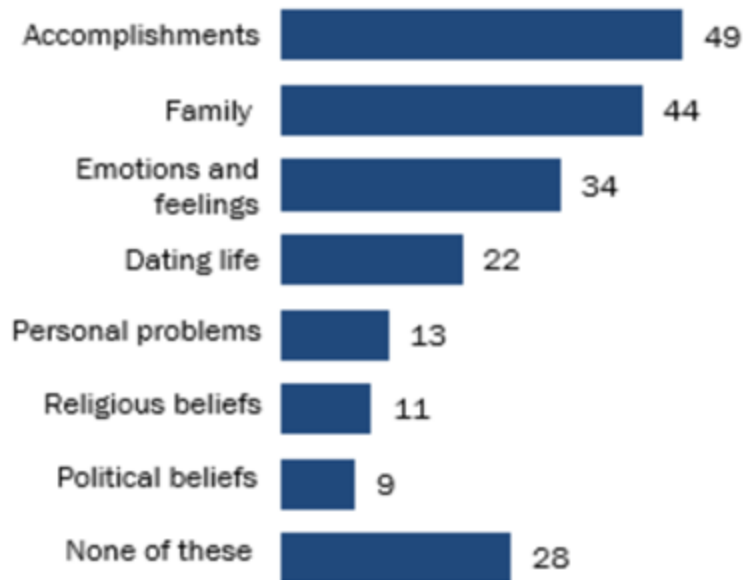
This means that percentages **should sum to 100%** (if using proportions) and counts should **sum to the total number of datapoints** (if using raw counts).

Let's see some examples.

Does this chart show a distribution?

While about half of teens post their accomplishments on social media, few discuss their religious or political beliefs

% of U.S. teens who say they ever post about their ___ on social media



Note: Respondents were allowed to select multiple options.

Respondents who did not give an answer are not shown.

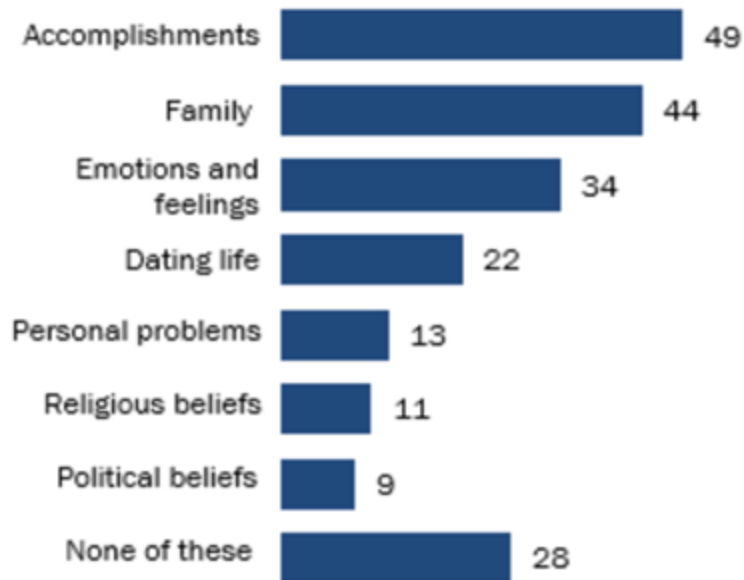
Source: Survey conducted March 7–April 10, 2018.

"Teens' Social Media Habits and Experiences"

Does this chart show a distribution?

While about half of teens post their accomplishments on social media, few discuss their religious or political beliefs

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"Teens' Social Media Habits and Experiences"

PEW RESEARCH CENTER

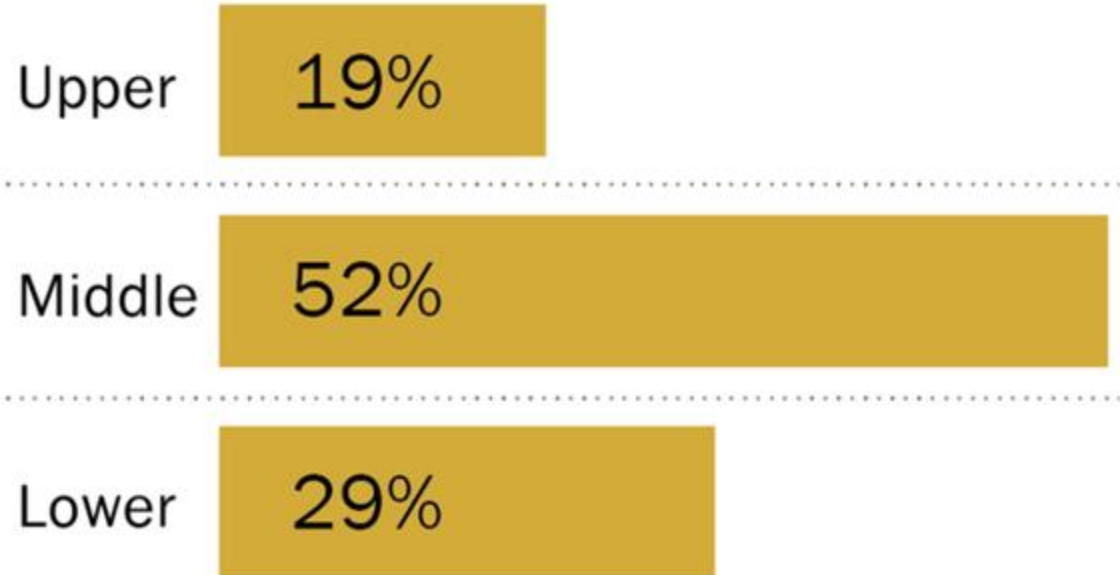
Does this chart show a distribution?

No.

- The chart does show percents of individuals in different categories!
- But, this is not a distribution because individuals can be in more than one category (see the fine print).

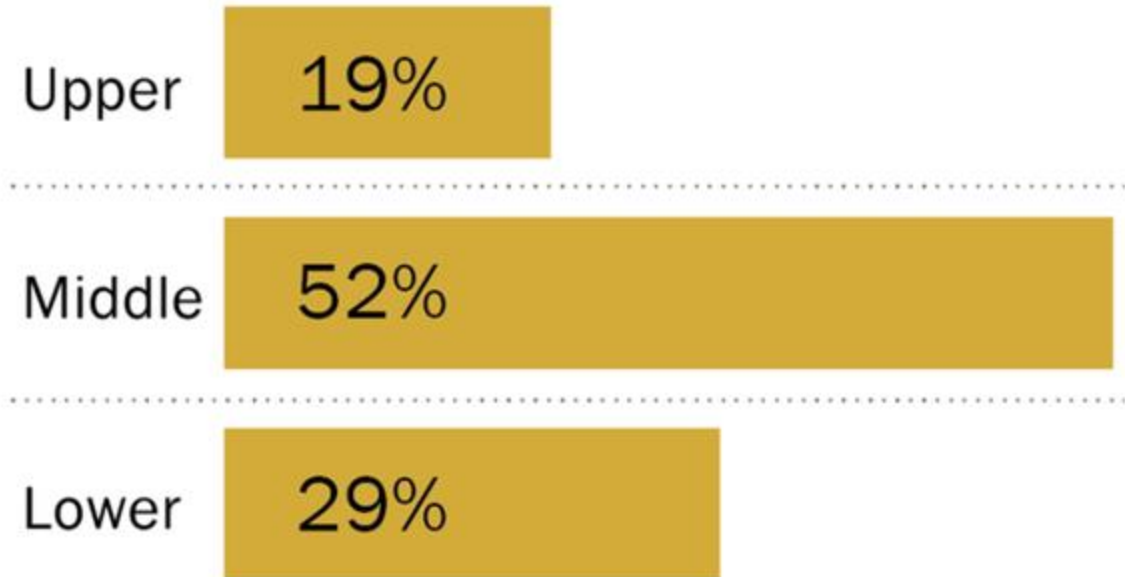
Does this chart show a distribution?

SHARE OF AMERICAN ADULTS
IN EACH INCOME TIER



Does this chart show a distribution?

SHARE OF AMERICAN ADULTS
IN EACH INCOME TIER



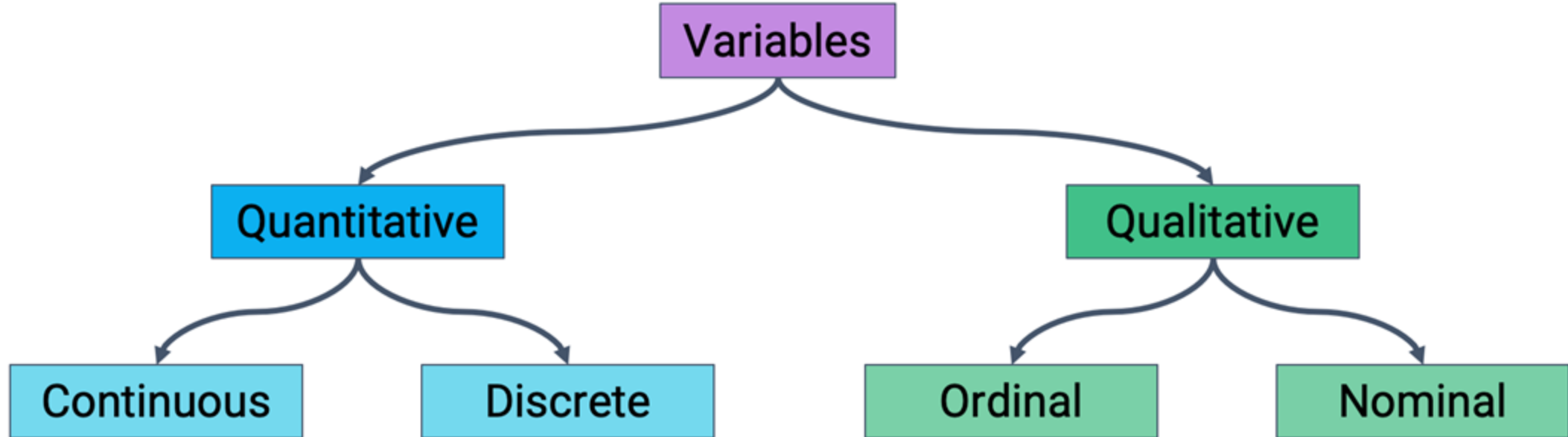
Does this chart show a distribution?

Yes!

- This chart shows the distribution of the qualitative ordinal variable "income tier."
- Each individual is in exactly one category.
- The values we see are the proportions of individuals in that category.
- Everyone is represented, as the total percentage is 100%.

Variable Types Should Inform Plot Choice

Different plots are more or less suited for displaying particular types of variables.

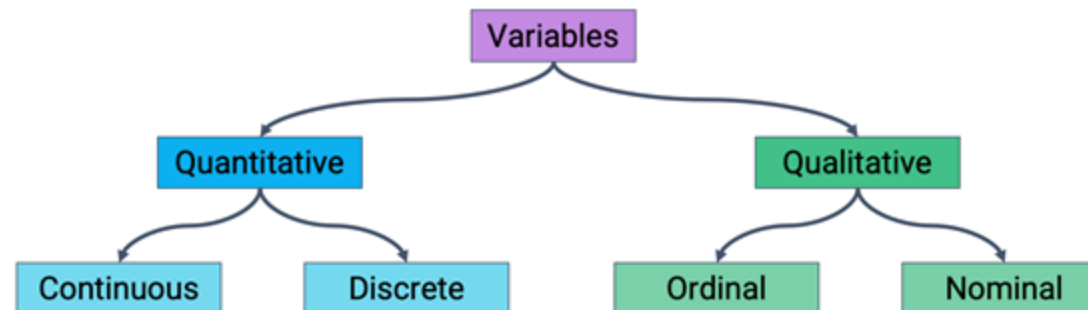


First step of visualization: Identify the variables being visualized. Then, select a plot type accordingly.

Question?

Usman wants to organize a picnic for all Data Science students to celebrate the end of the semester but can't decide which type of soda to get. They decide to consult the SCIT staff about their favorite fizzy drinks.

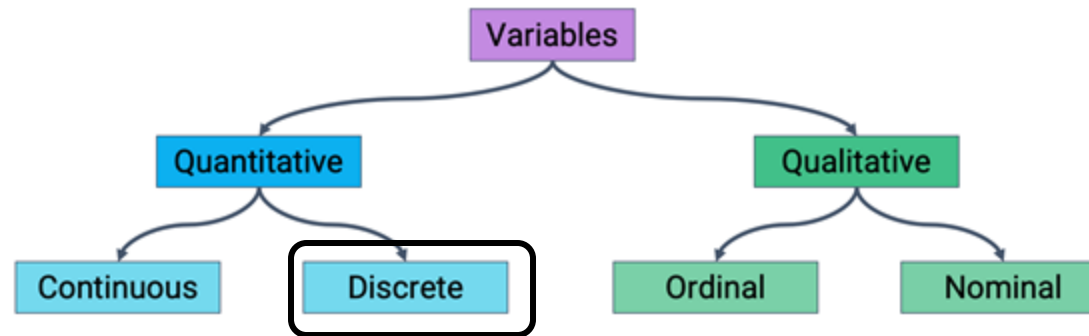
	Name	Soda	Falvor	Year
0	Huda	Diet Coke	Cola	Graduate
1	Amna	Orange Fanta	Orange	5
2	Shazia	Coca-cola	Cola	4
3	Nouman	Mountain Dew	Lemon-Lime	3
4	Asim	Sprite	Lemon-Lime	3
5	Nimra	None	None	3
6	Hamna	A & W	Mint	3



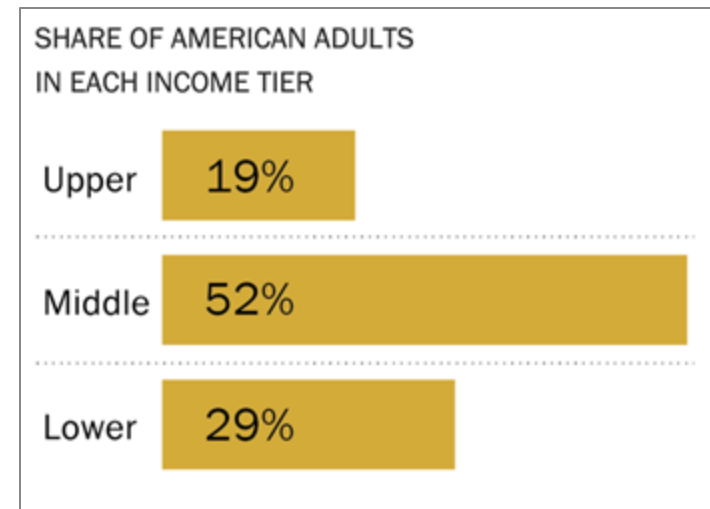
Bar Plots: Distributions of Qualitative Variables

Bar plots are the most common way of displaying the **distribution** of a **qualitative** variable.

*Sometimes quantitative discrete data too, if there are few unique values.



- For example, the proportion of adults in the upper, middle, and lower classes.
- Lengths encode values.
 - *Widths* encode *nothing*!
 - *Color* could indicate a sub-category (but not necessarily).



World Bank Dataset

We will be using the `wb` dataset about world countries for most of our work today.

	Continent	Country	Primary completion rate: Male: % of relevant age group: 2015	Primary completion rate: Female: % of relevant age group: 2015	Lower secondary completion rate: Male: % of relevant age group: 2015	Lower secondary completion rate: Female: % of relevant age group: 2015	Youth literacy rate: Male: % of ages 15-24: 2005-14	Youth literacy rate: Female: % of ages 15-24: 2005-14	Adult literacy rate: Male: % ages 15 and older: 2005-14	Adult literacy rate: Female: % ages 15 and older: 2005-14
0	Africa	Algeria	106.0	105.0	68.0	85.0	96.0	92.0	83.0	68.0
1	Africa	Angola	NaN	NaN	NaN	NaN	79.0	67.0	82.0	60.0
2	Africa	Benin	83.0	73.0	50.0	37.0	55.0	31.0	41.0	18.0
3	Africa	Botswana	98.0	101.0	86.0	87.0	96.0	99.0	87.0	89.0
5	Africa	Burundi	58.0	66.0	35.0	30.0	90.0	88.0	89.0	85.0

Generating Bar Plots: Matplotlib

We will mainly use two libraries for generating plots: [Matplotlib](#) and [Seaborn](#).

Most Matplotlib plotting functions follow the same structure: We pass in a sequence (**list**, **array**, or **Series**) of values to be plotted on the x-axis, and a second sequence of values to be plotted on the y-axis.

```
import matplotlib.pyplot as plt  
plt.plotting_function(x_values, y_values)
```

Matplotlib is typically given the alias `plt`



To add labels and a title:

```
plt.xlabel("x axis label")  
plt.ylabel("y axis label")  
plt.title("Title of the plot");
```

Generating Bar Plots: Matplotlib

To create a bar plot in Matplotlib: `plt.bar( )`

[\[Documentation\]](#)

```
continents = wb["Continent"].value_counts()
```

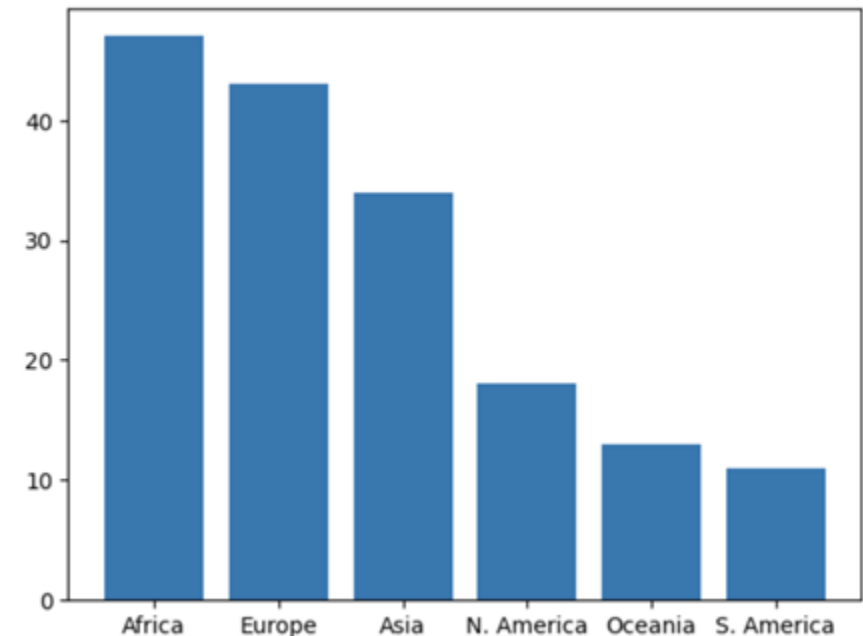
```
plt.bar(continents.index, continents.values);
```

x values

y values

Africa	47
Europe	43
Asia	34
N. America	18
Oceania	13
S. America	11

Name: Continent, dtype: int64

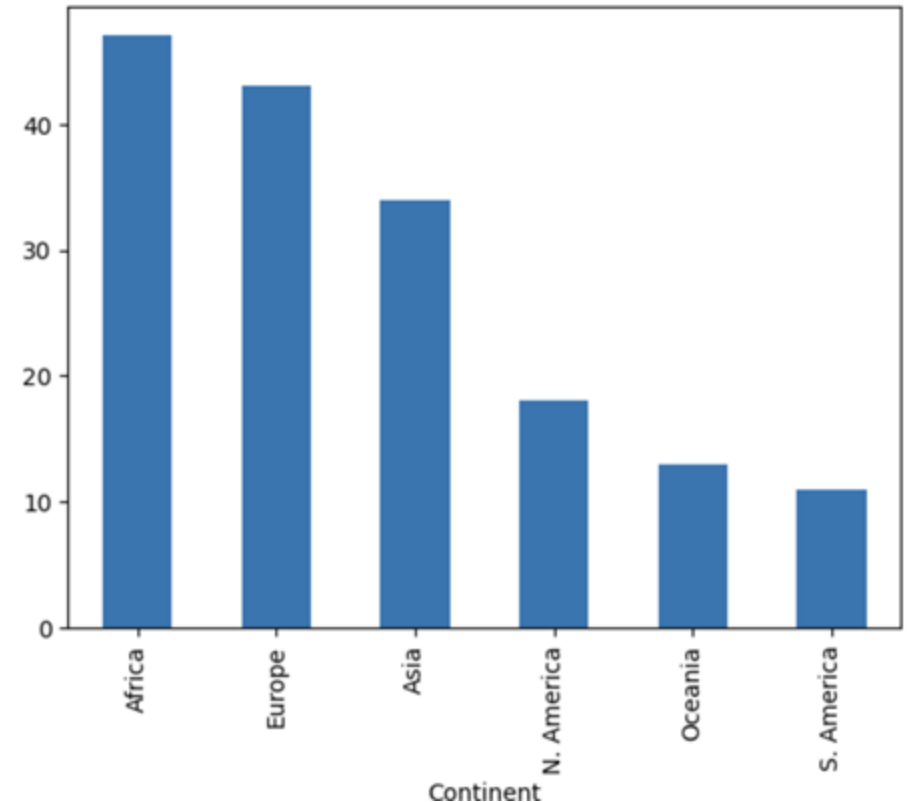


Generating Bar Plots: pandas Native Plotting

To create a bar plot in native pandas: `.plot(kind='bar')`

```
Africa      47
Europe      43
Asia        34
N. America  18
Oceania     13
S. America  11
Name: Continent, dtype: int64
```

```
wb["Continent"].value_counts().plot(kind='bar')
```



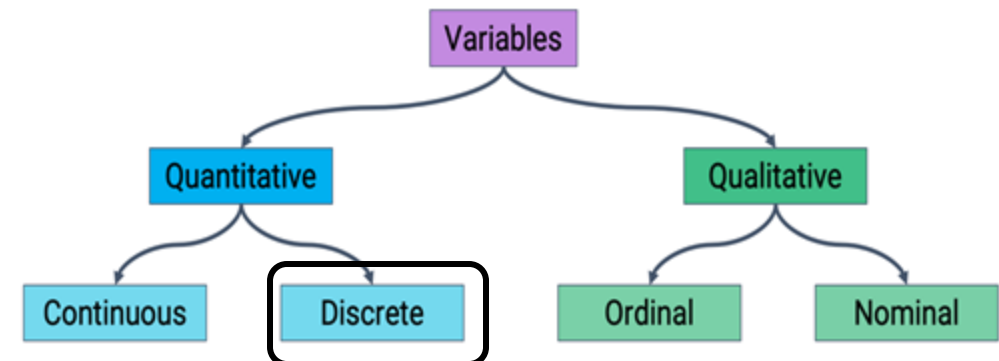
Generating Bar Plots: Seaborn

Seaborn plotting functions use a different structure: Pass in an entire **DataFrame**, then specify what column(s) to plot.

```
import seaborn as sns ← Seaborn is typically given the alias sns  
sns.plotting_function(data=df, x="x_col", y="y_col")  
sns.plotting_function(data=df, x="x_col")  
sns.plotting_function(data=df, y="y_col")
```

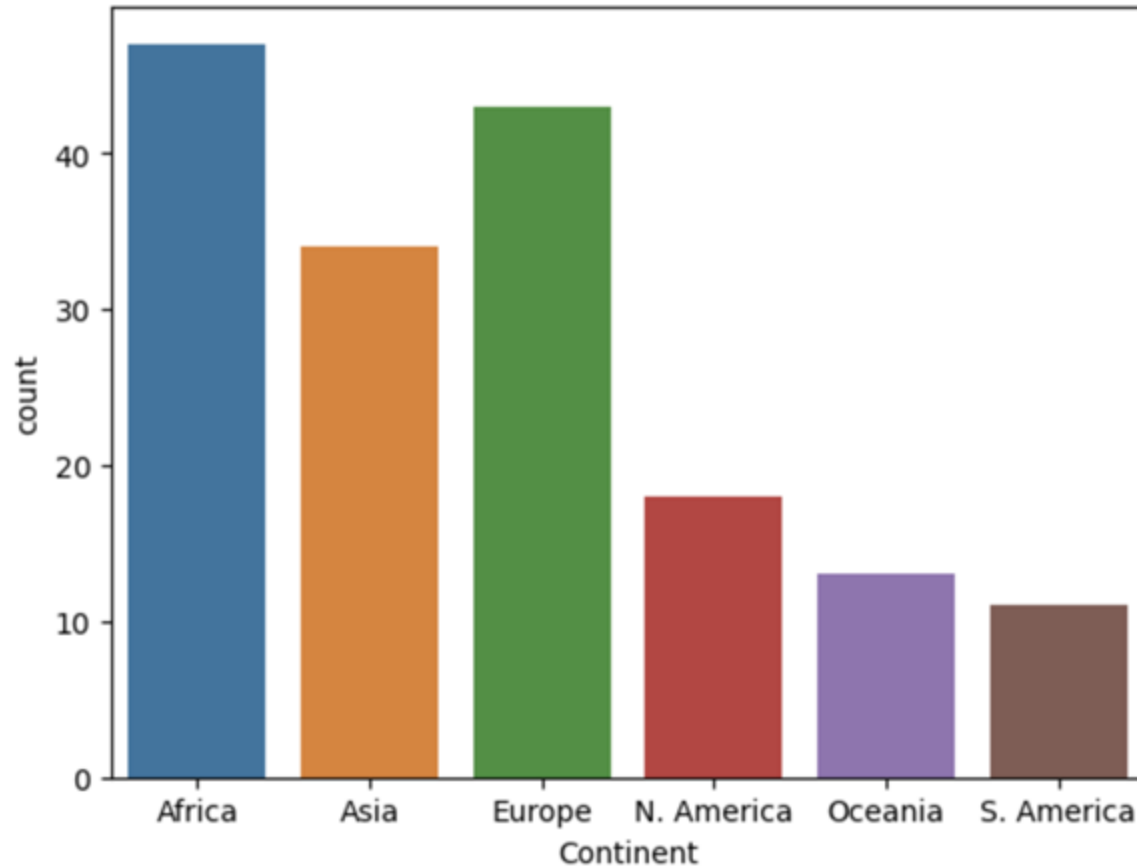
To add labels and a title, use the same syntax as before:

```
plt.xlabel("x axis label")  
plt.ylabel("y axis label")  
plt.title("Title of the plot");
```



To create a bar plot in Seaborn: `sns.countplot( )`

[\[Documentation\]](#)



`countplot` operates at a higher level of abstraction!

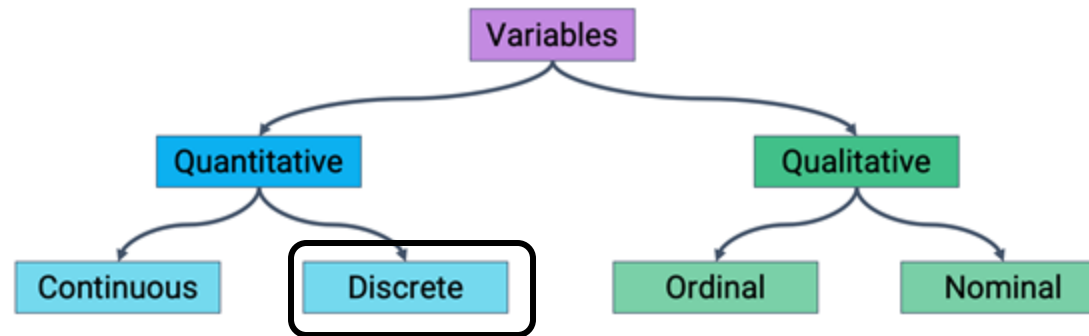
You give it the entire **DataFrame** and it does the counting for you.

```
import seaborn as sns  
sns.countplot(data=wb, x="Continent");
```

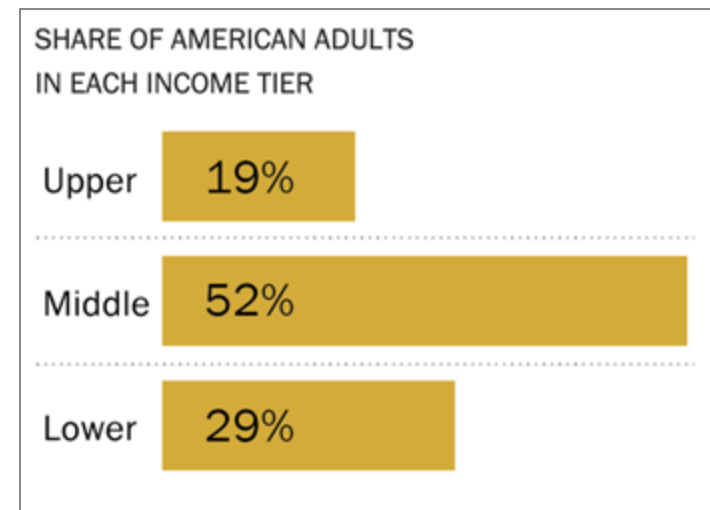
Bar Plots: Distributions of Qualitative Variables

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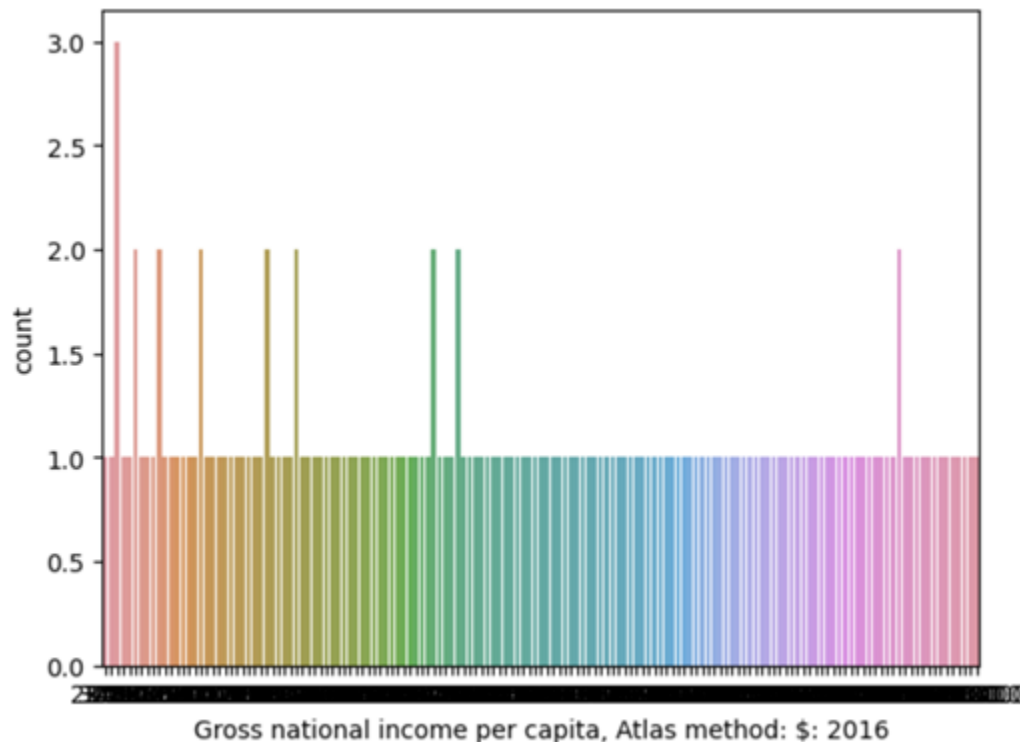


Distributions of Quantitative Variables

Earlier, we said that bar plots are appropriate for distributions of qualitative variables.

Why only qualitative? Why not quantitative as well?

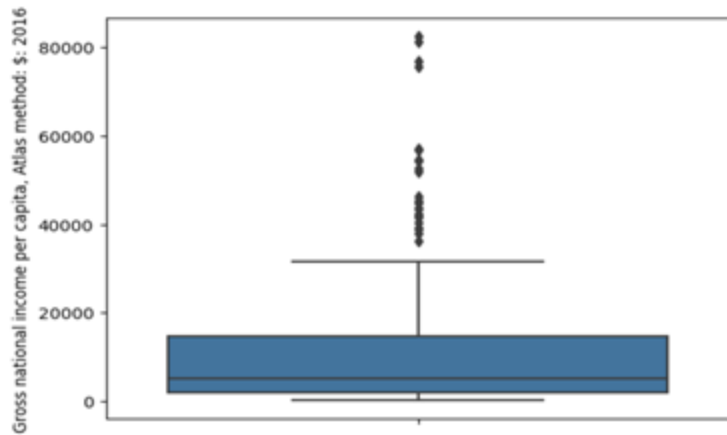
- For example: The distribution of gross national income per capita.



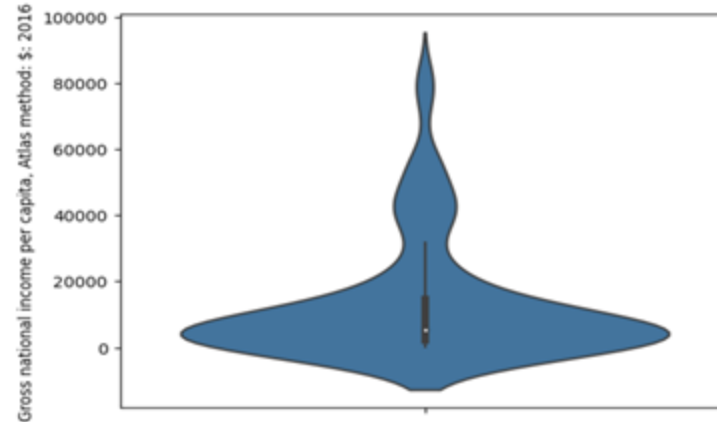
A bar plot will create a separate bar for each unique value. This leads to too many bars for continuous data!

Distributions of Quantitative Variables

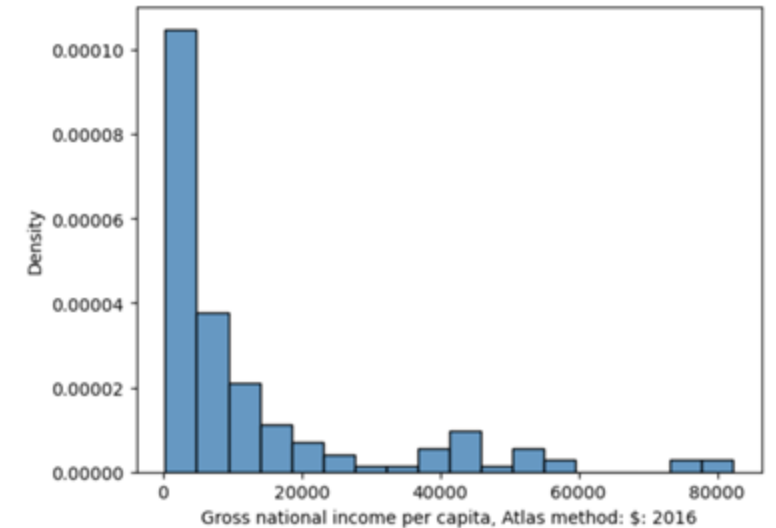
To visualize the distribution of a continuous quantitative variable:



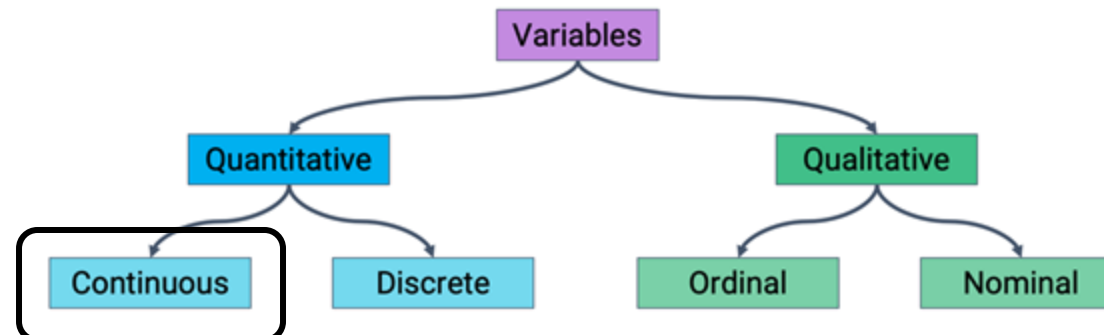
Box plot



Violin plot



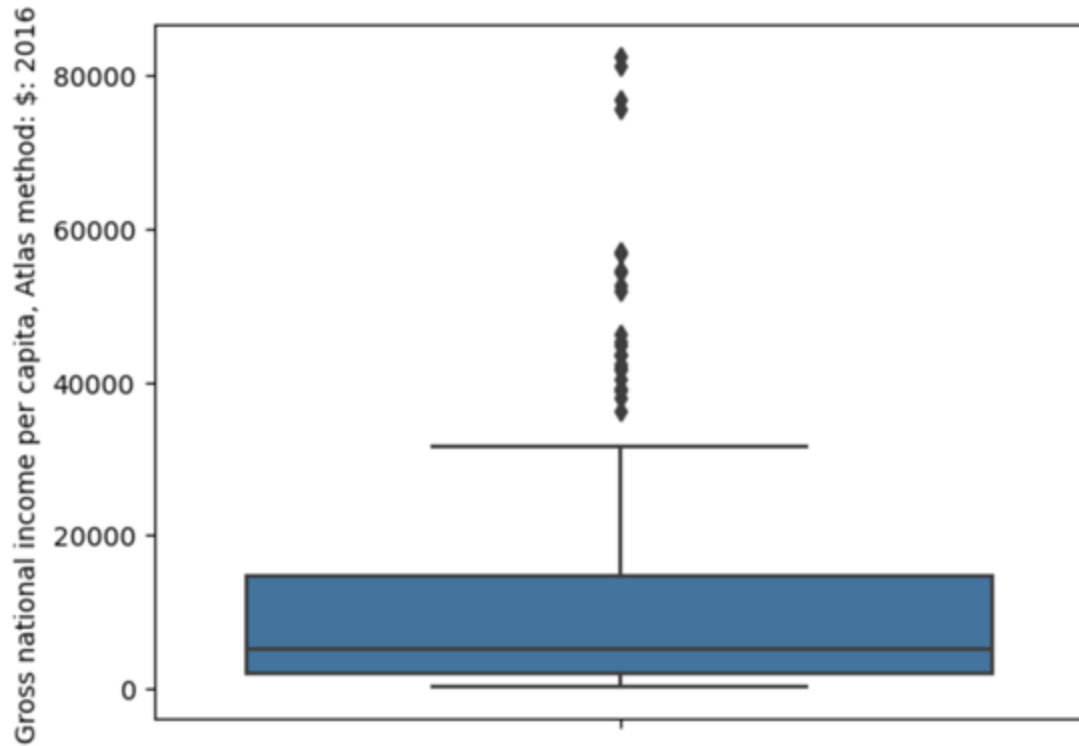
Histogram



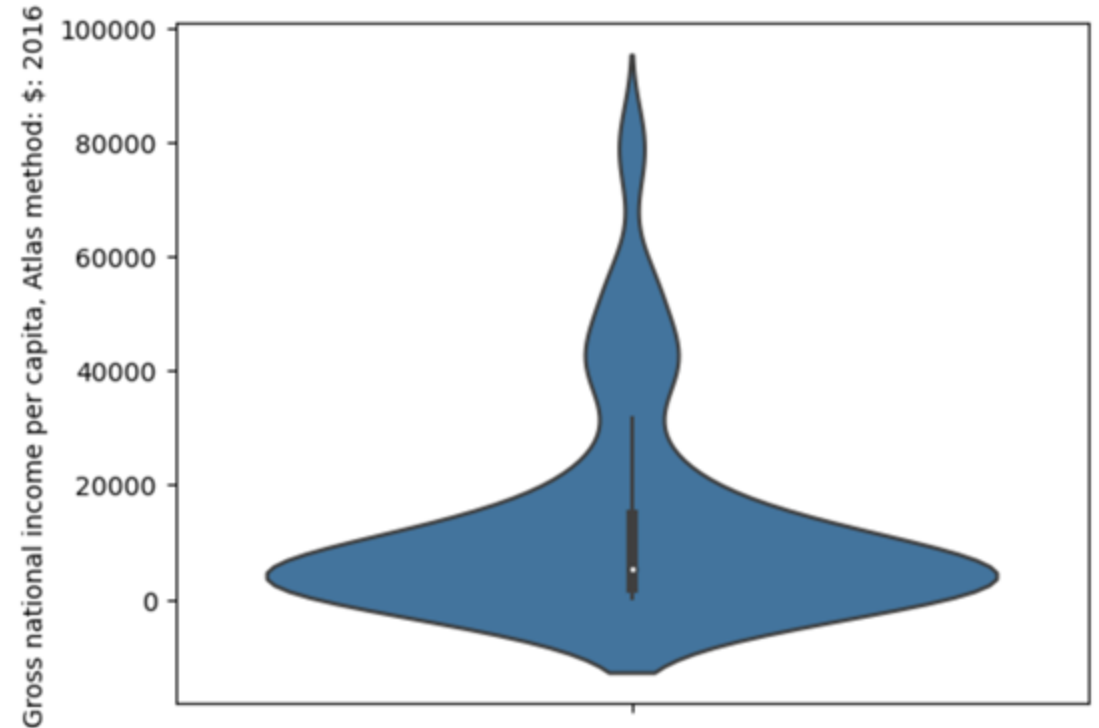
Box plots and Violin Plots

Box plots and violin plots display distributions using information about **quartiles**.

- In a box plot, the width of the box encodes no meaning.
- In a violin plot, the width of the "violin" indicates the density of datapoints at each value.



```
sns.boxplot(data=df, y="y_variable");  
\[Documentation\]
```



```
sns.violinplot(data=df, y="y_variable");  
\[Documentation\]
```

Quartiles

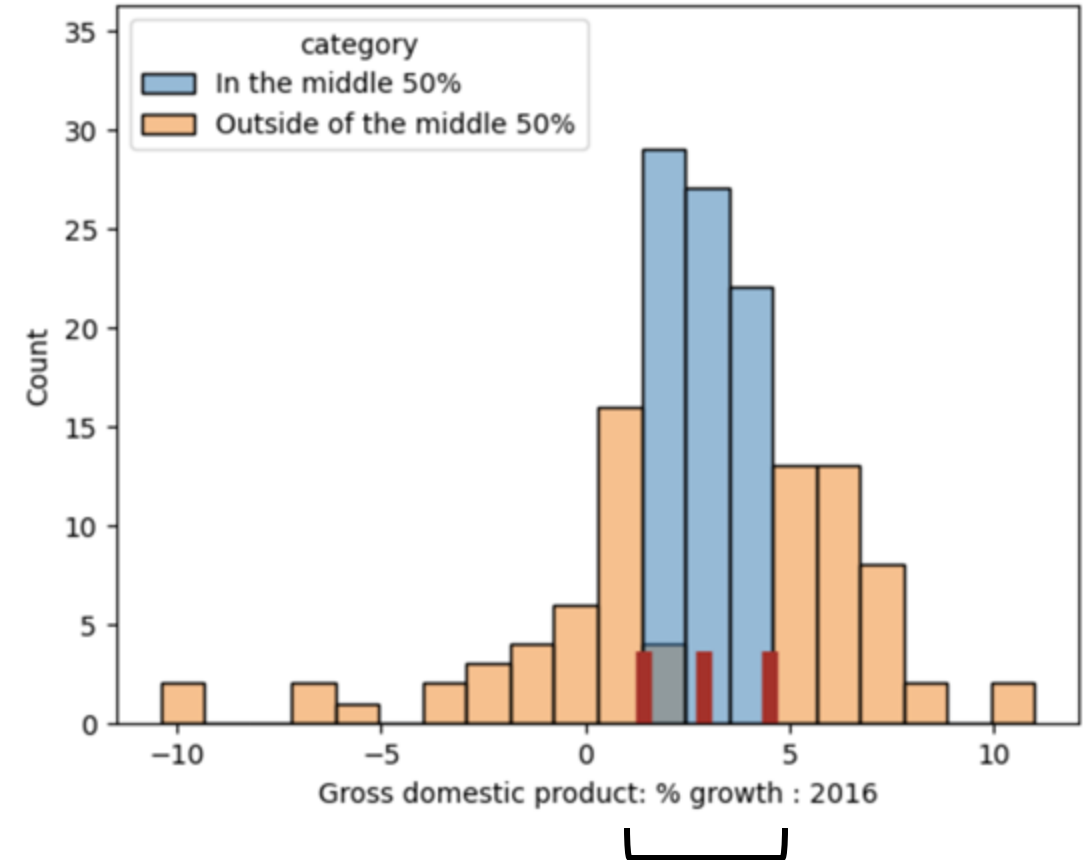
For a quantitative variable:

- First or lower quartile: 25th percentile.
- Second quartile: 50th percentile (median).
- Third or upper quartile: 75th percentile.

The interval [first quartile, third quartile] contains the "middle 50%" of the data.

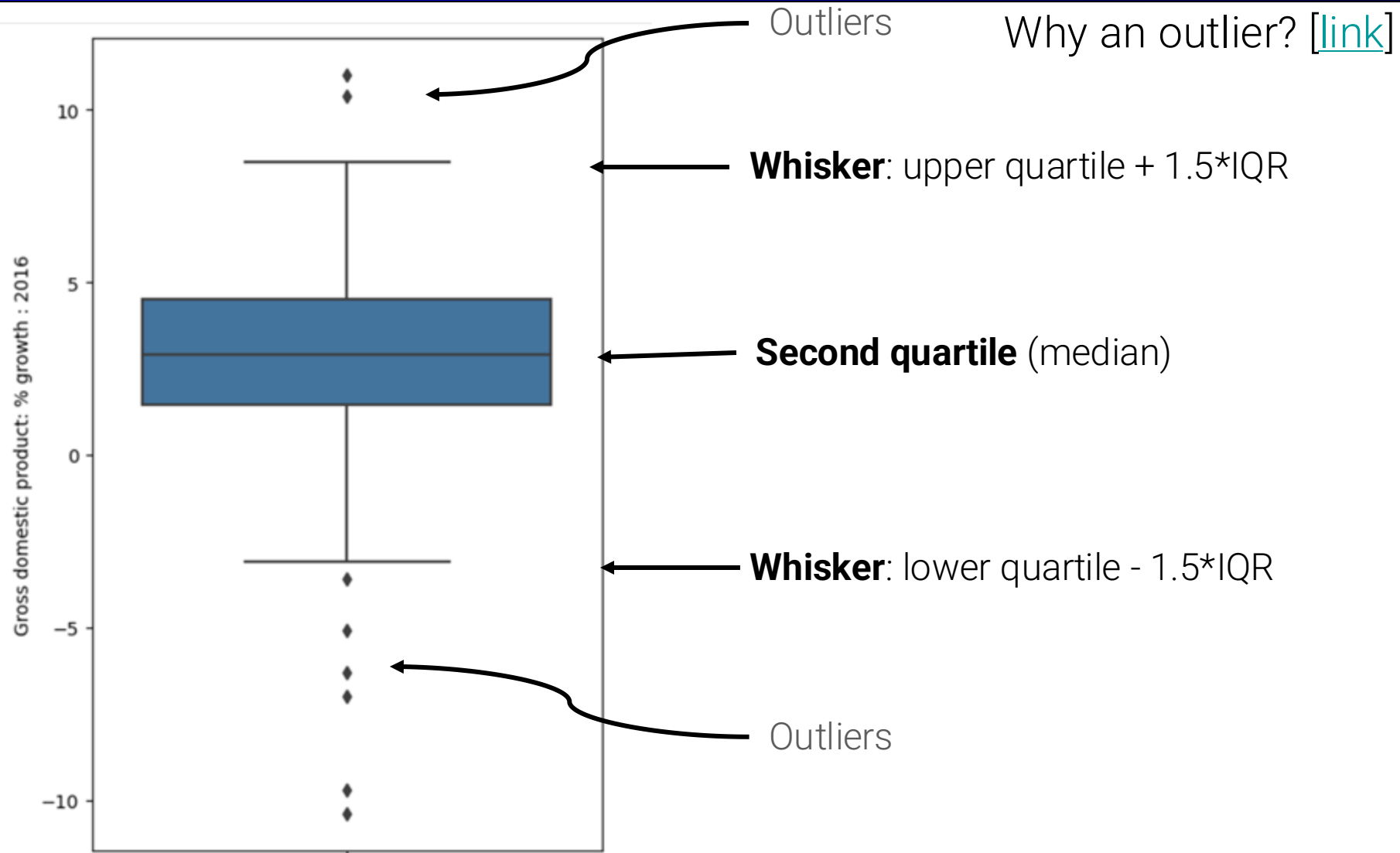
Interquartile range (IQR) measures spread.

- $IQR = \text{third quartile} - \text{first quartile}$.



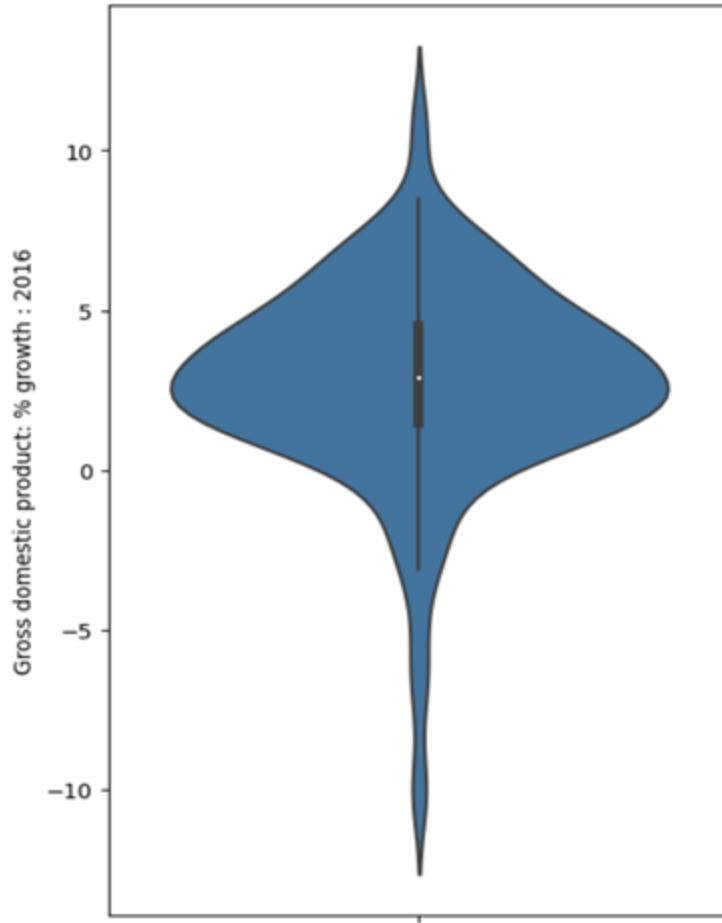
The length of this region is the IQR

Box Plots



```
sns.boxplot(data=wb, y="Gross domestic product: % growth : 2016")
```

Violin Plots



Violin plots are similar to box plots, but also show smoothed density curves.

- The "width" of our "box" now has meaning!
- The three quartiles and "whiskers" are still present – look closely.

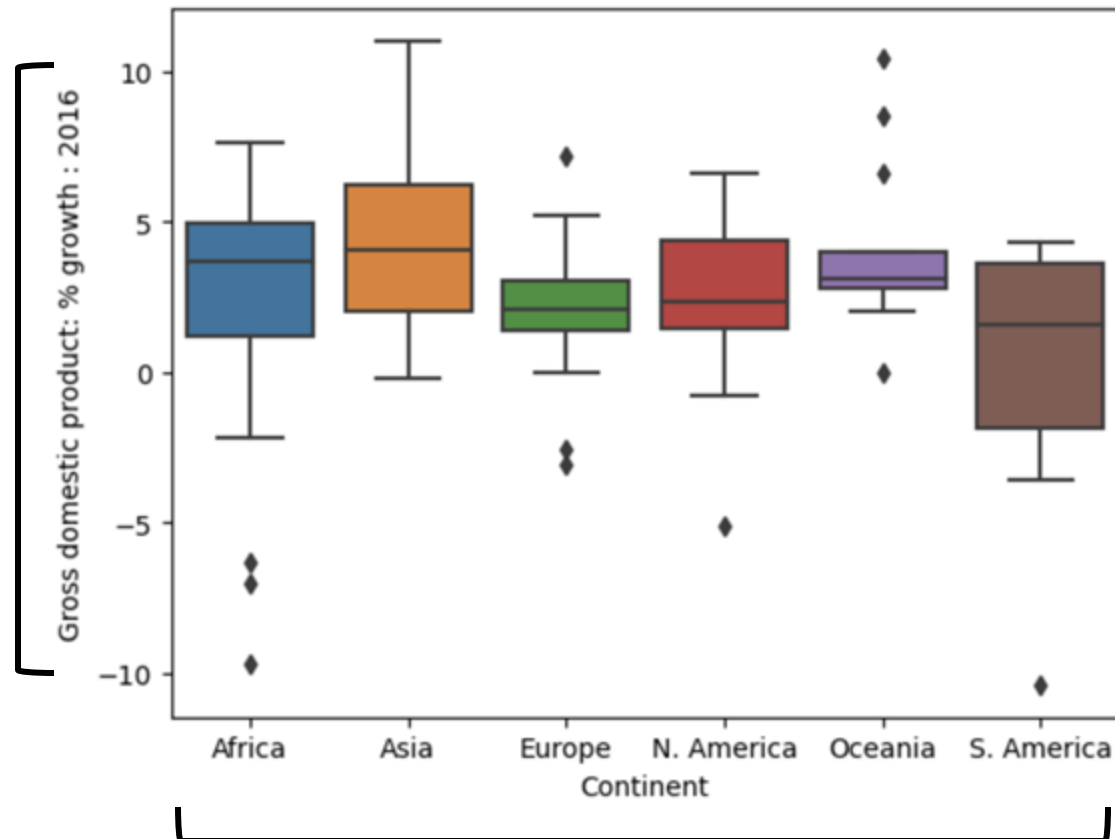
```
sns.violinplot(data=wb, y="Gross domestic product: % growth : 2016")
```

Side-by-side Box and Violin Plots

What if we wanted to incorporate a *qualitative* variable as well? For example, compare the distribution of a quantitative continuous variable *across* different qualitative categories.

```
sns.boxplot(data=wb, x="Continent", y="Gross domestic product: % growth : 2016");
```

GDP growth:
quantitative continuous

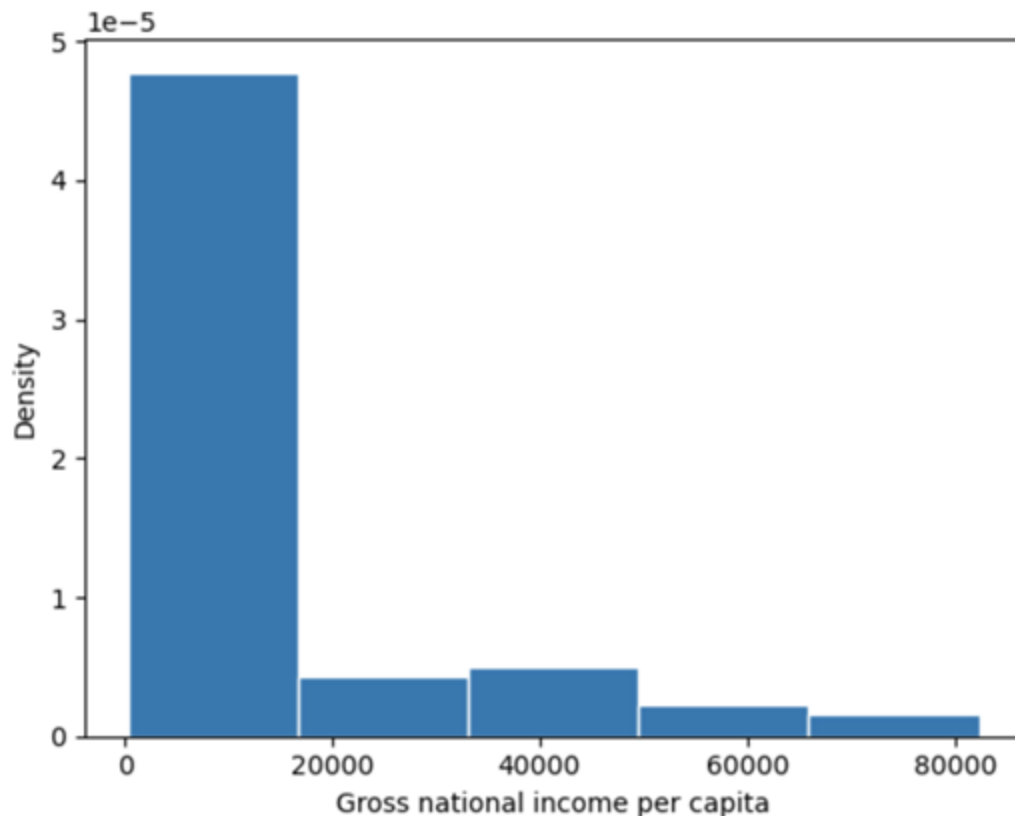


Continent: qualitative nominal

Histograms

A histogram:

- Collects datapoints with similar values into a shared "bin".
- Scales the bins such that the **area** of each bin is equal to the **percentage** of datapoints it contains.



The first bin has a width of \$16410
height of 4.77×10^{-5}

This means that it contains $16410 \times (4.77 \times 10^{-5}) = 78.3\%$ of all datapoints in the dataset.

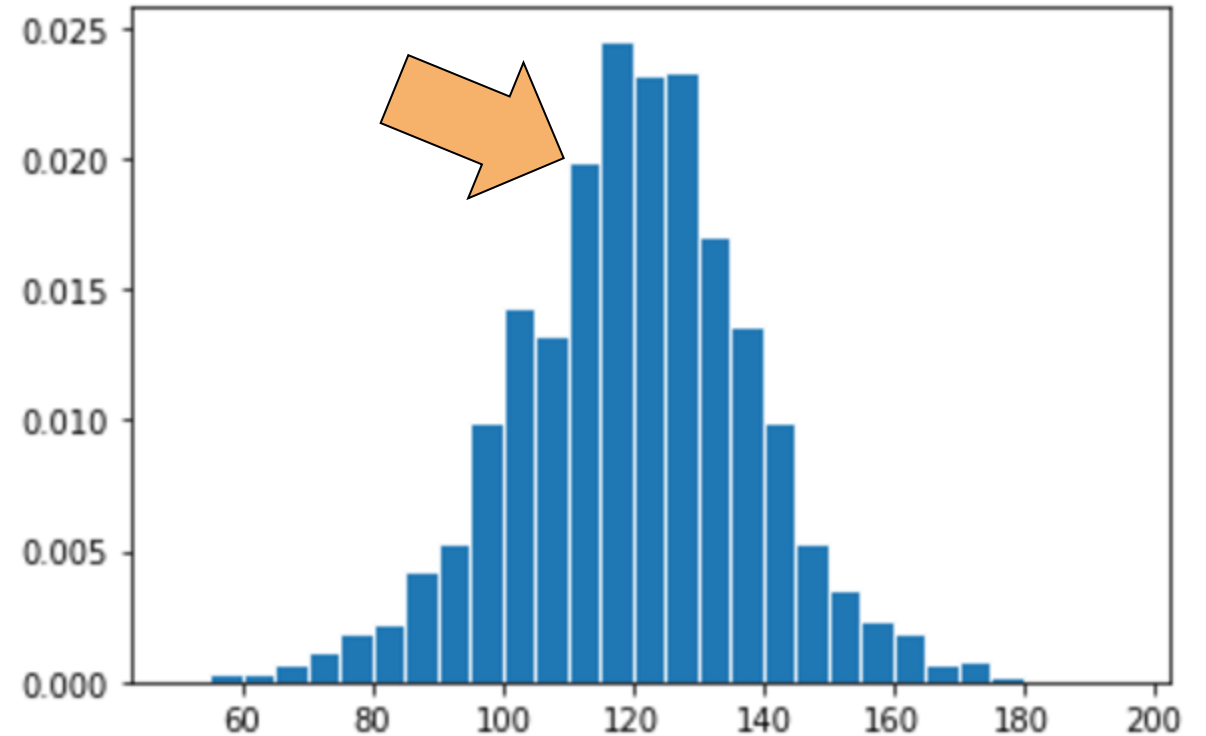
Question

How many observations are in the bin $[110, 115)$ given the following information?

- There are 1174 observations in total.**
- Width of bin $[110, 115)$: 5**
- Height of bar $[110, 115)$: 0.02**

There are 1174 observations total.

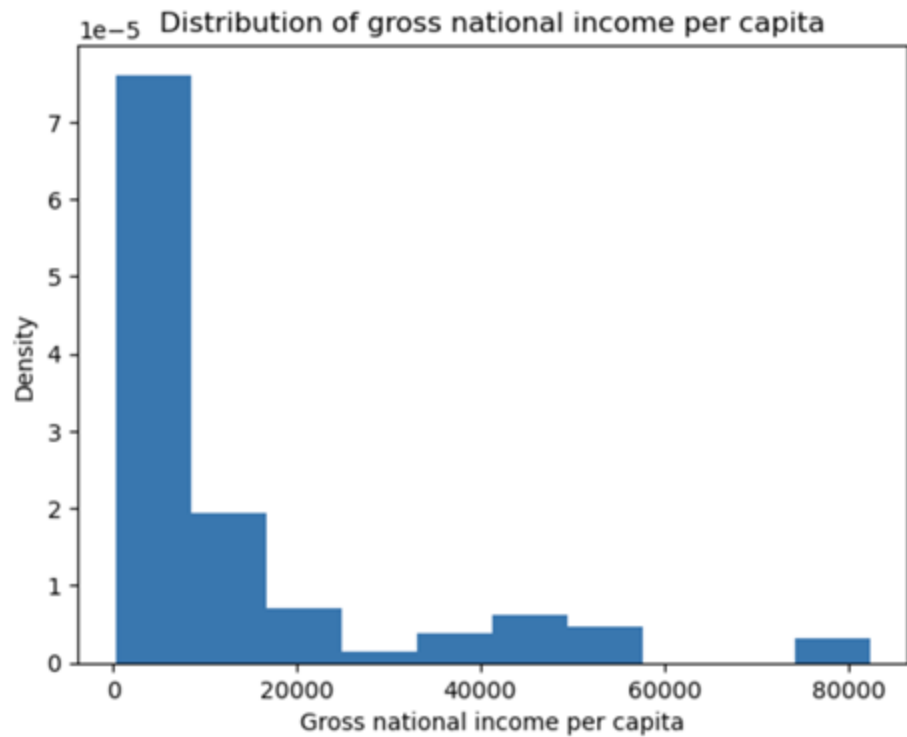
- Width of bin $[110, 115)$: 5
- Height of bar $[110, 115)$: 0.02
- Proportion in bin $= 5 * 0.02 = 0.1$
- Number in bin $= 0.1 * 1174 = \mathbf{117.4}$



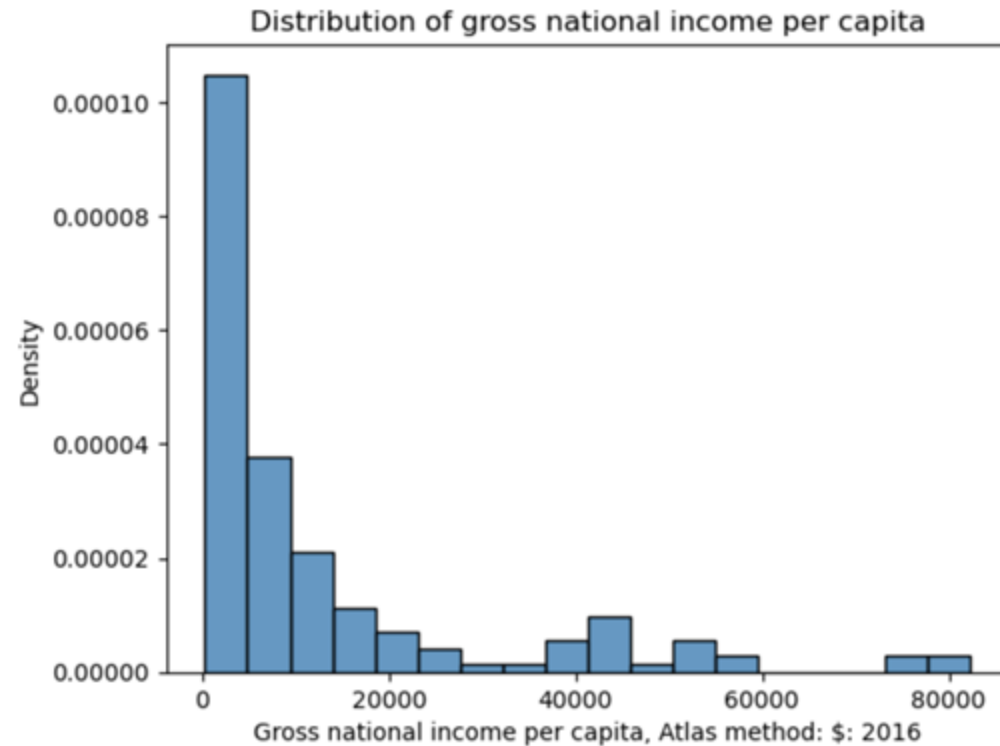
Histograms in Code

In Matplotlib [[Documentation](#)]: `plt.hist(x_values, density=True)`

In Seaborn [[Documentation](#)]: `sns.histplot(data=df, x="x_column", stat="density")`



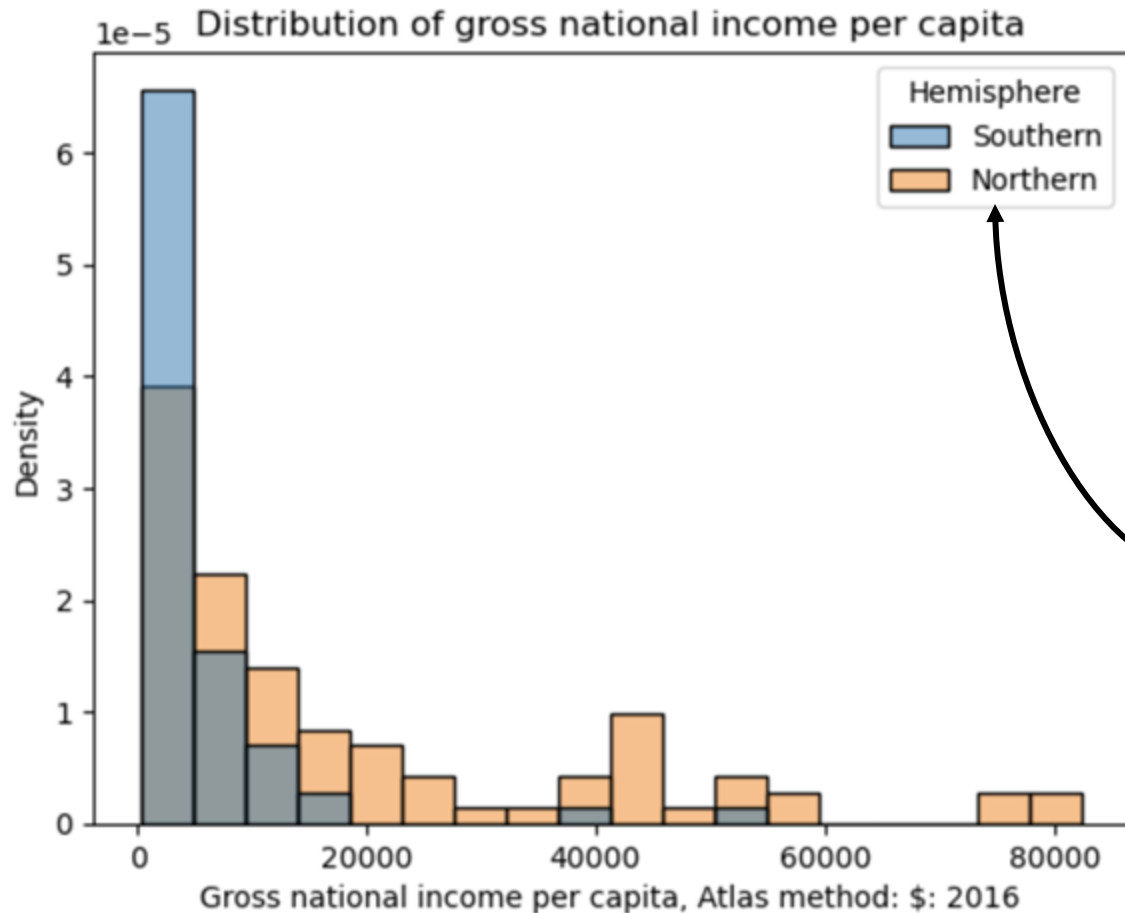
Matplotlib



Seaborn

Overlaid Histograms

To compare a quantitative variable's distribution across qualitative categories, overlay histograms on top of one another.



The **hue** parameter of Seaborn plotting functions sets the column that should be used to determine color.

```
sns.histplot(data=wb, hue="Hemisphere",  
x="Gross national income...")
```

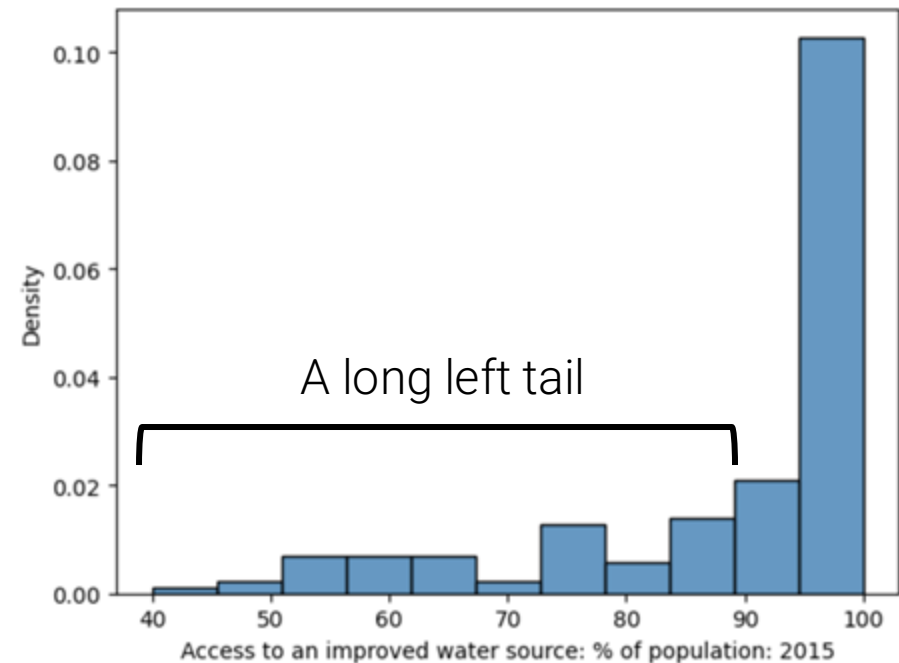
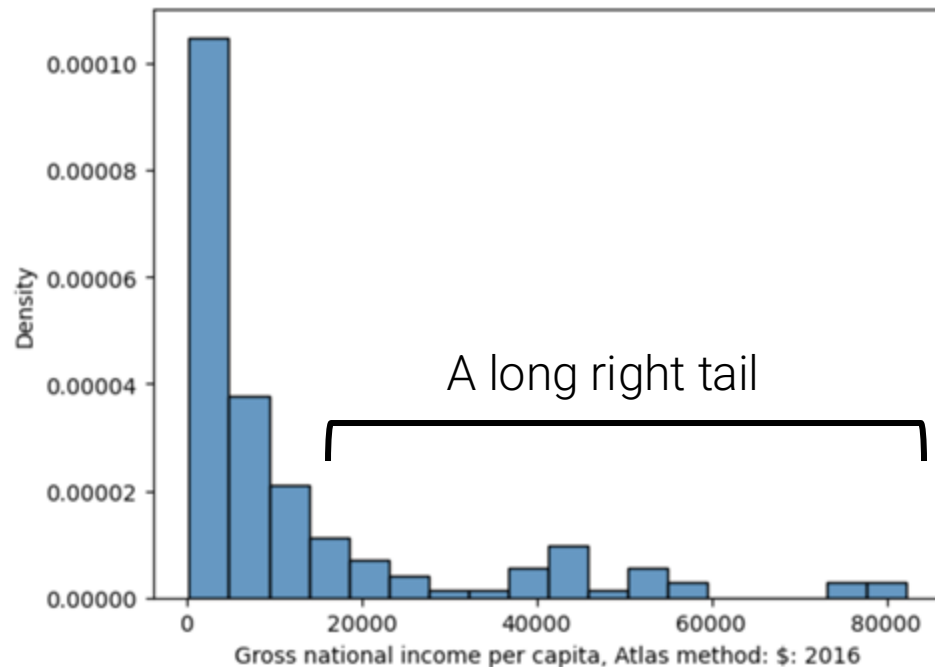
Always include a legend when color is used to encode information!

Interpreting Histograms

The **skew** of a histogram describes the direction in which its "tail" extends.

- A distribution with a long right tail is skewed right.
- A distribution with a long left tail is skewed left.

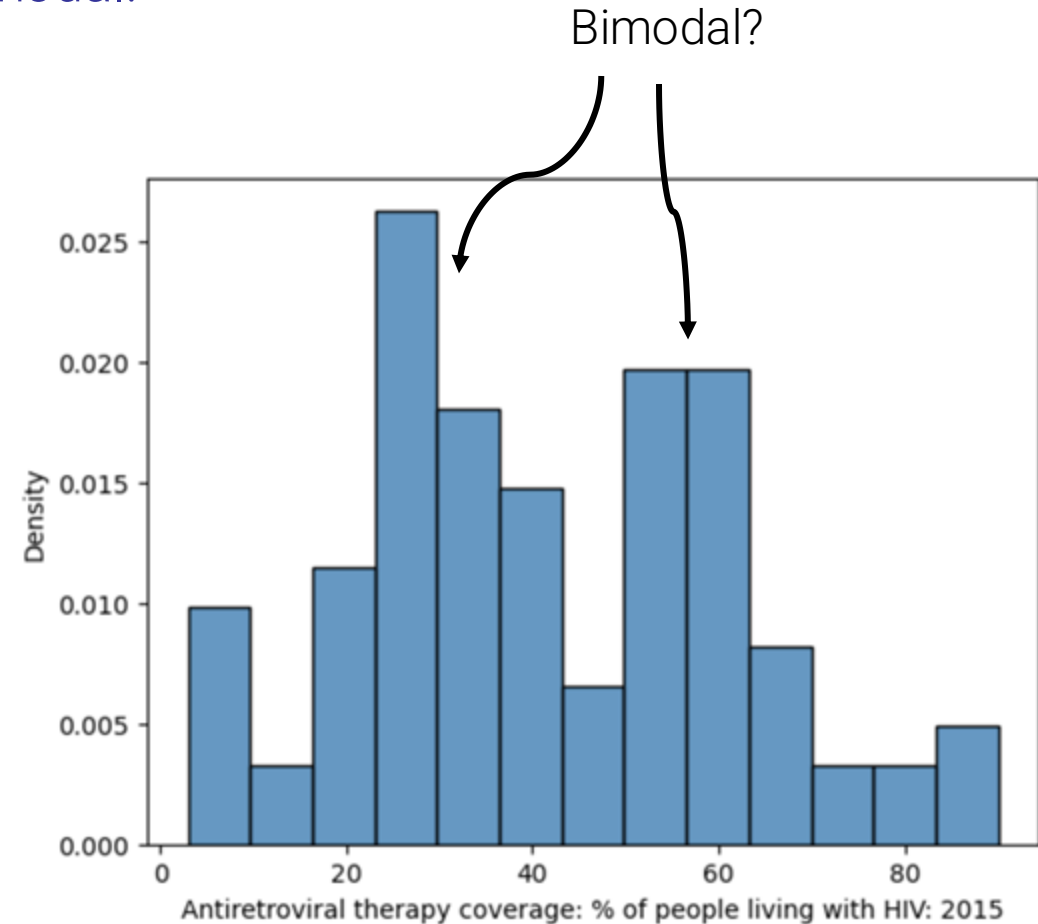
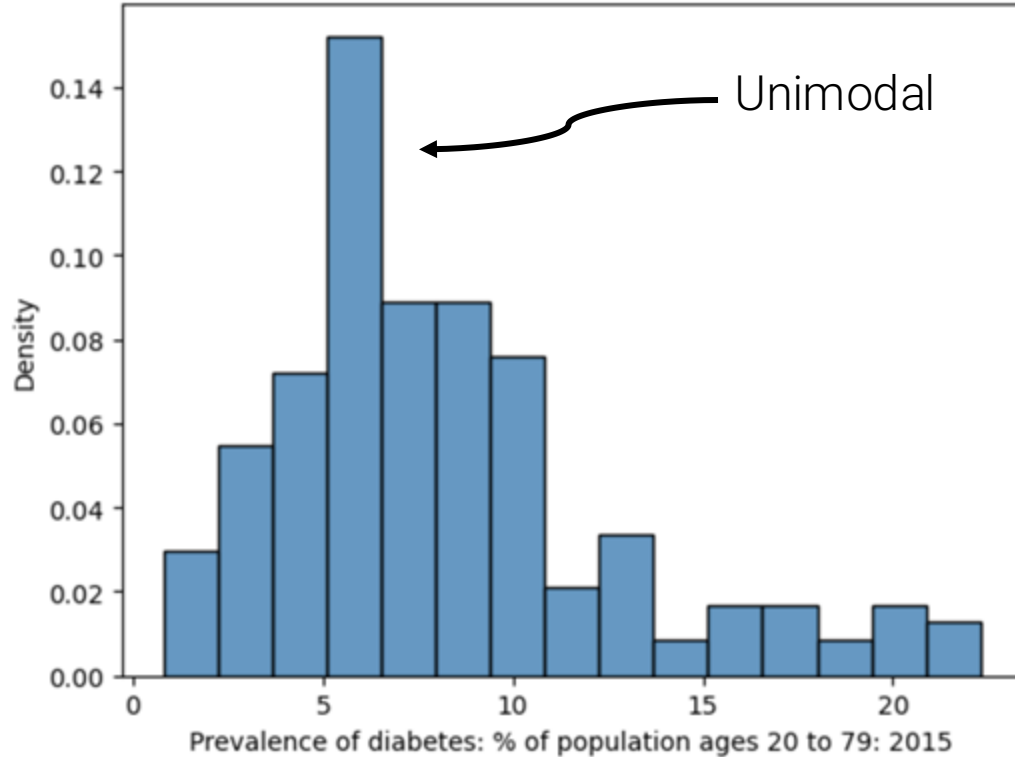
A histogram with no clear skew is called symmetric.



Interpreting Histograms

The **mode(s)** of a histogram are the peak values in the distribution.

- A distribution with one clear peak is called unimodal.
- Two peaks: bimodal.
- More peaks: multimodal.



Visualization Lecture ended here



Made by DALL-E

Kernel Density Estimation

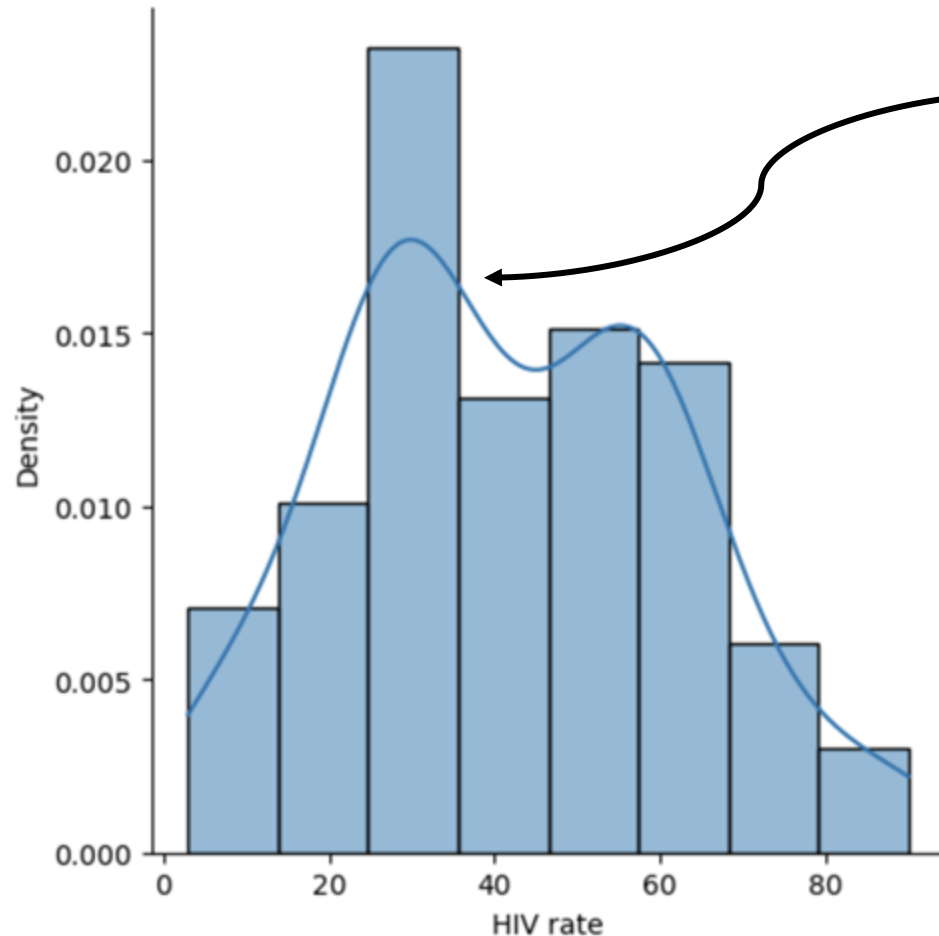
Lecture 7, Data 100 Spring 2024

- Goals of visualization
- Visualizing distributions
- **Kernel density estimation**

Kernel Density Estimation: Intuition

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r
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General trends across a distribution,
Smoothing a distribution helps
the data and eliminate noise.

Idea: approximate the probability distribution that generated the data.

- Assign an “error range” to each data point in the dataset – if we were to sample the data again, we might get a different value.
- Sum up the error ranges of all data points.
- Scale the resulting distribution to integrate to 1.

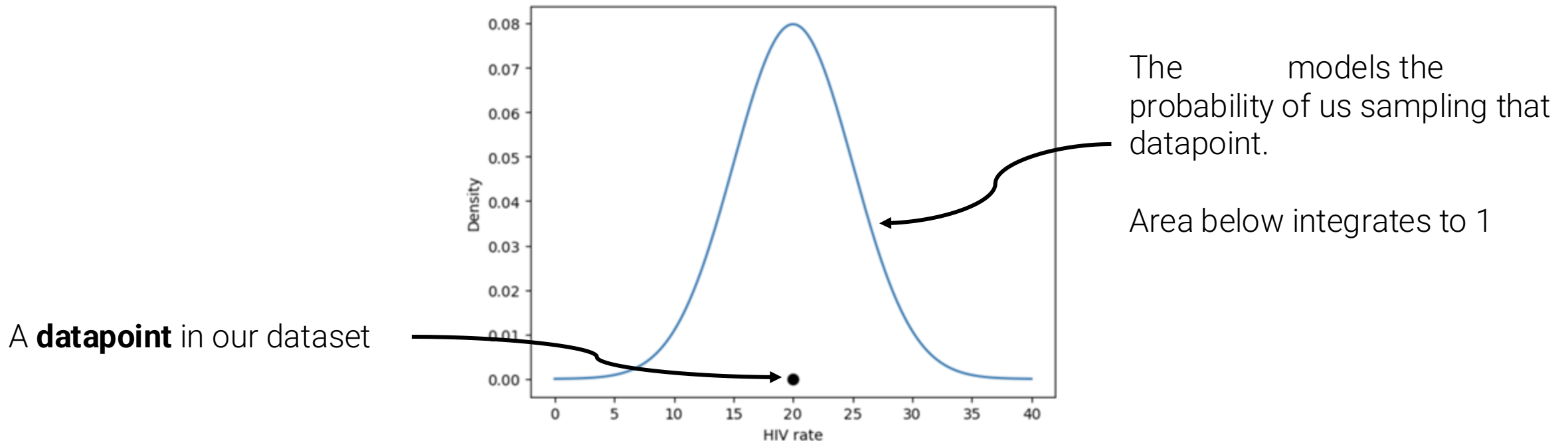
Kernel Density Estimation: Process

Idea: Approximate the probability distribution that generated the data.

1308228

- Place a kernel at each data point.
- Normalize kernels so that total area = 1.
- Sum all kernels together.

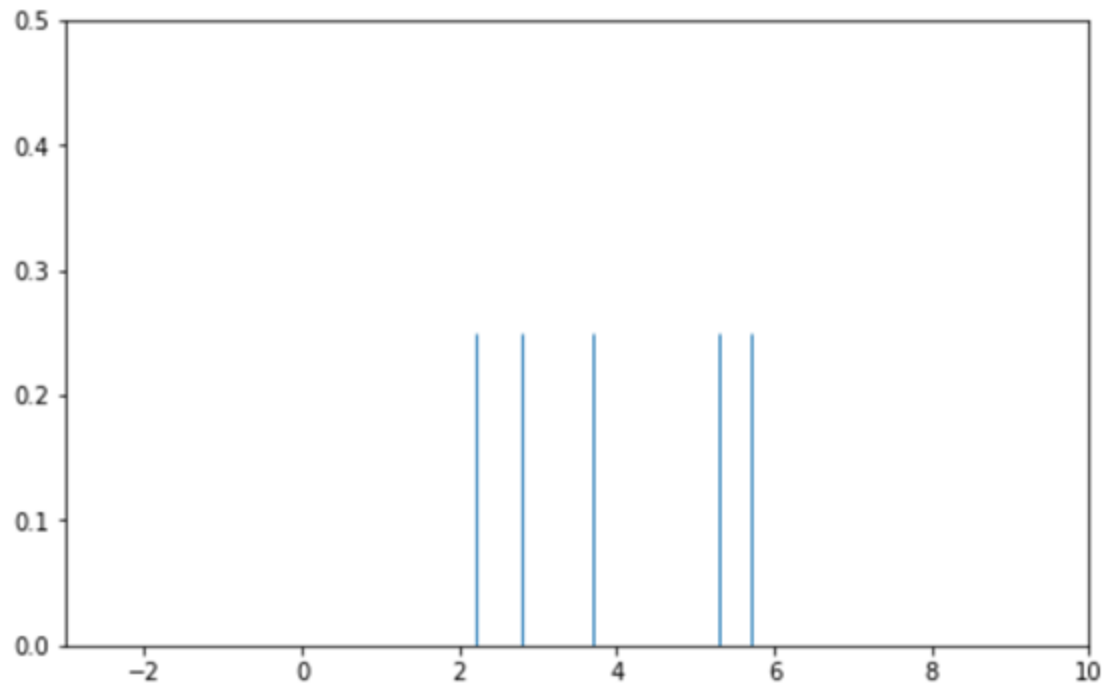
A **kernel** is a function that tries to capture the randomness of our sampled data.



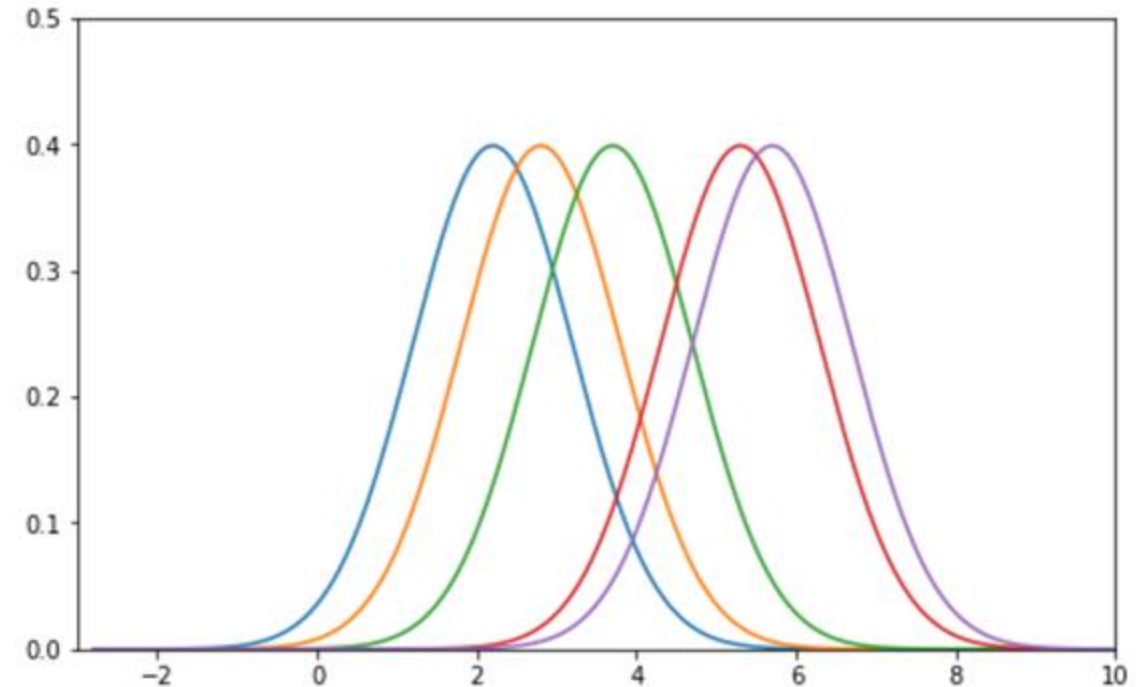
Step 1 – Place a Kernel at Each Data Point

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Consider a fake dataset with just five collected datapoints.



Each line represents a datapoint in the dataset (e.g. one country's HIV rate).



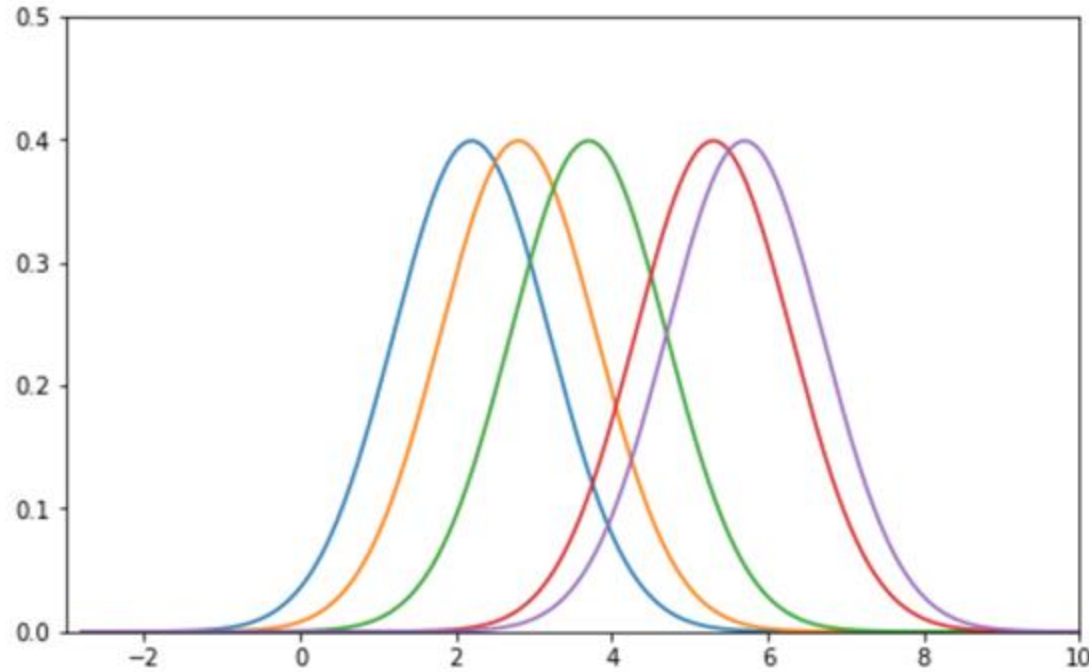
Place a kernel on top of each datapoint.

Step 2 – Normalize Kernels

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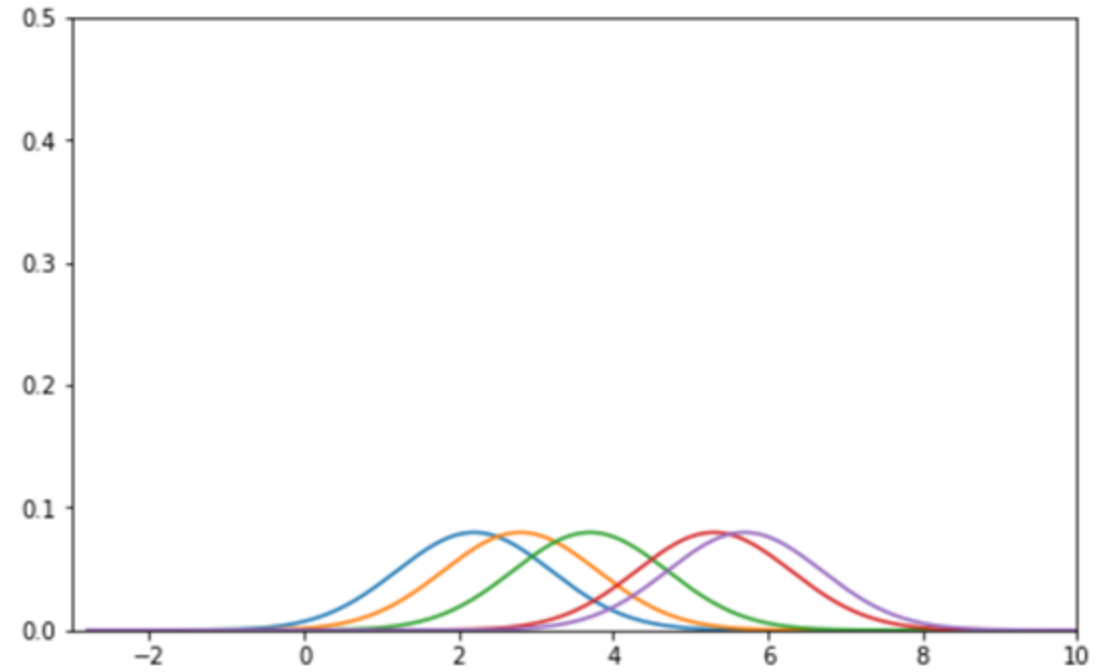
In Step 3, We will be summing each of these kernels to produce a probability distribution

-
-
-



Each kernel has area 1.

C
 $\lambda(x)$
1/5

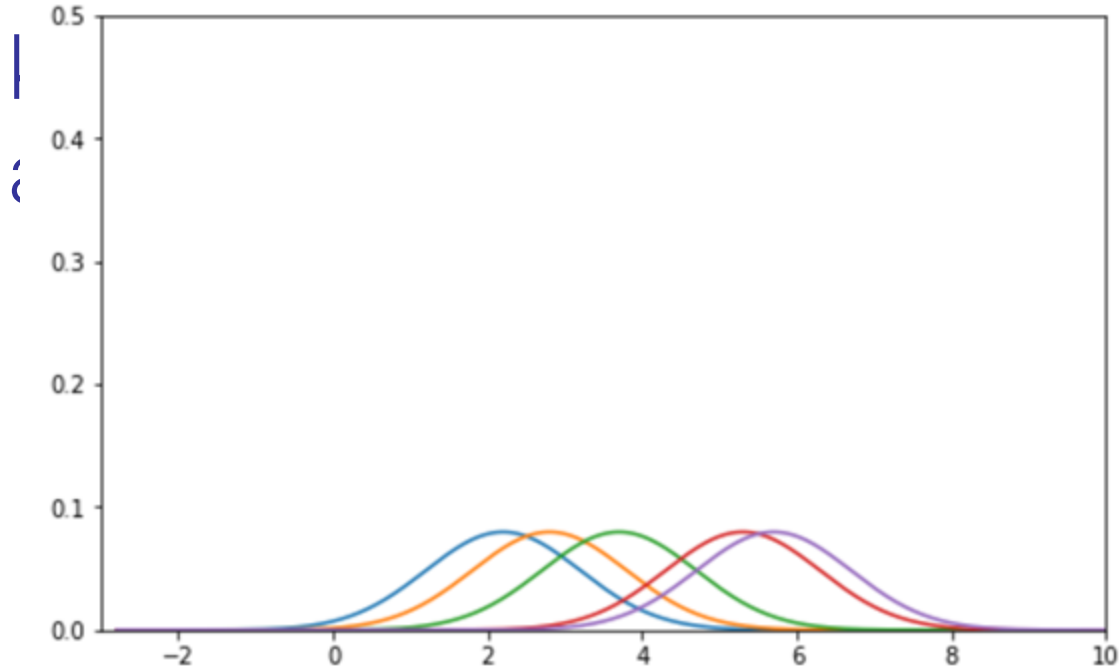


Each normalized kernel has density $\frac{1}{5}$.

Step 3 – Sum the Normalized Kernels

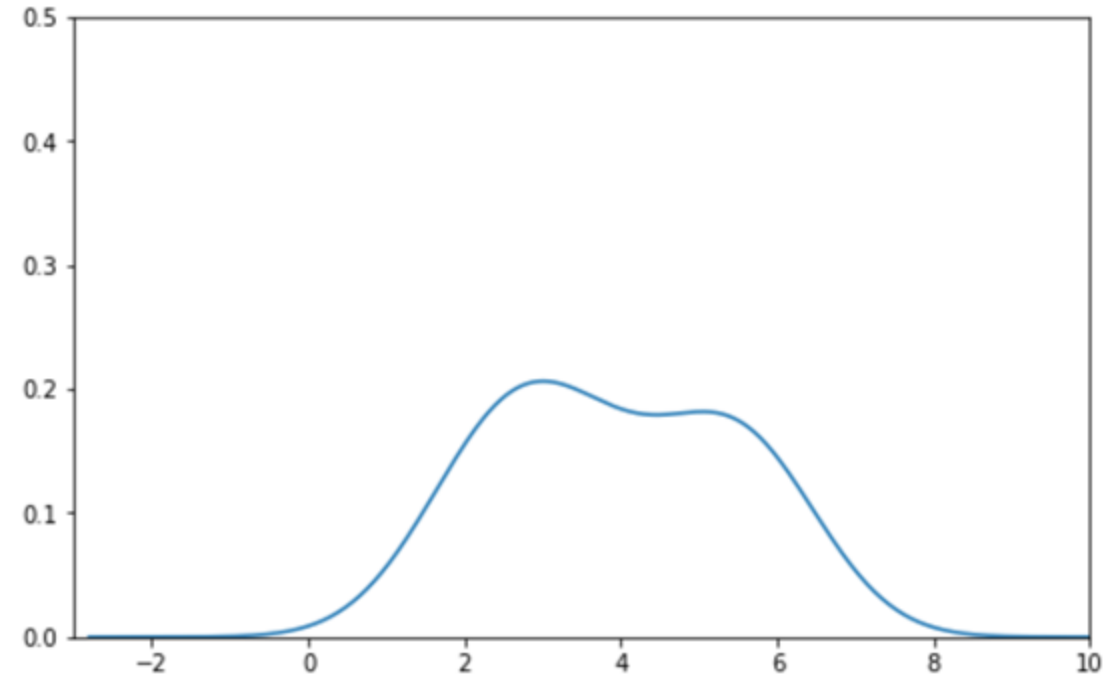
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At each point in the distribution, add up the values of all



Sum these five normalized curves together.

h c
dist

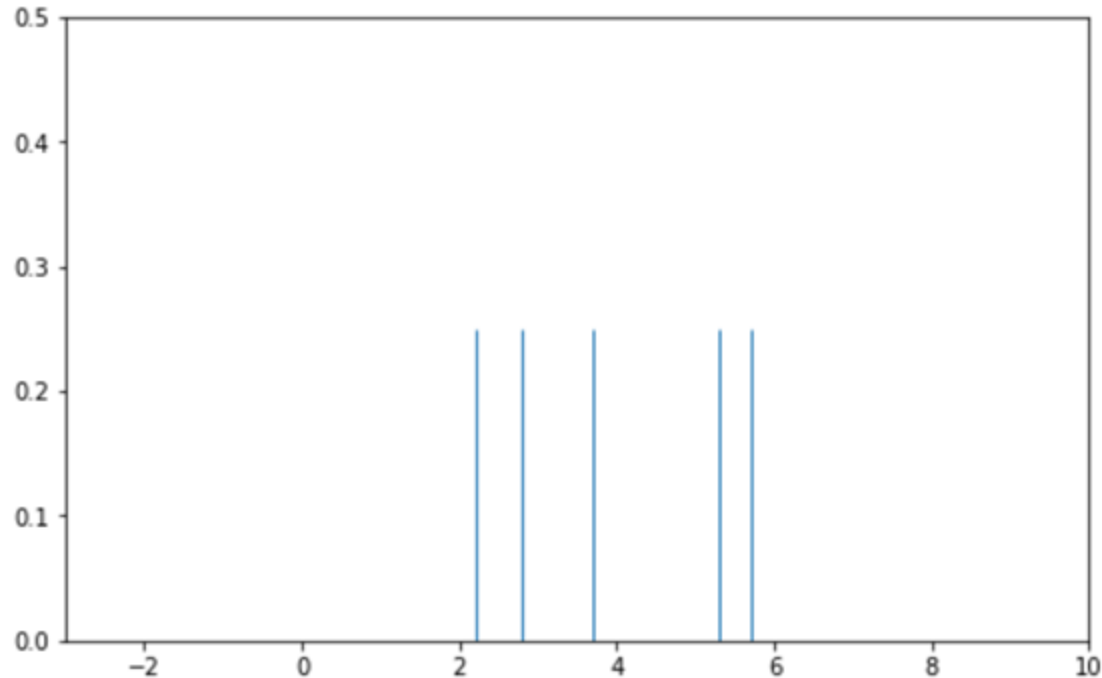


The final KDE curve.

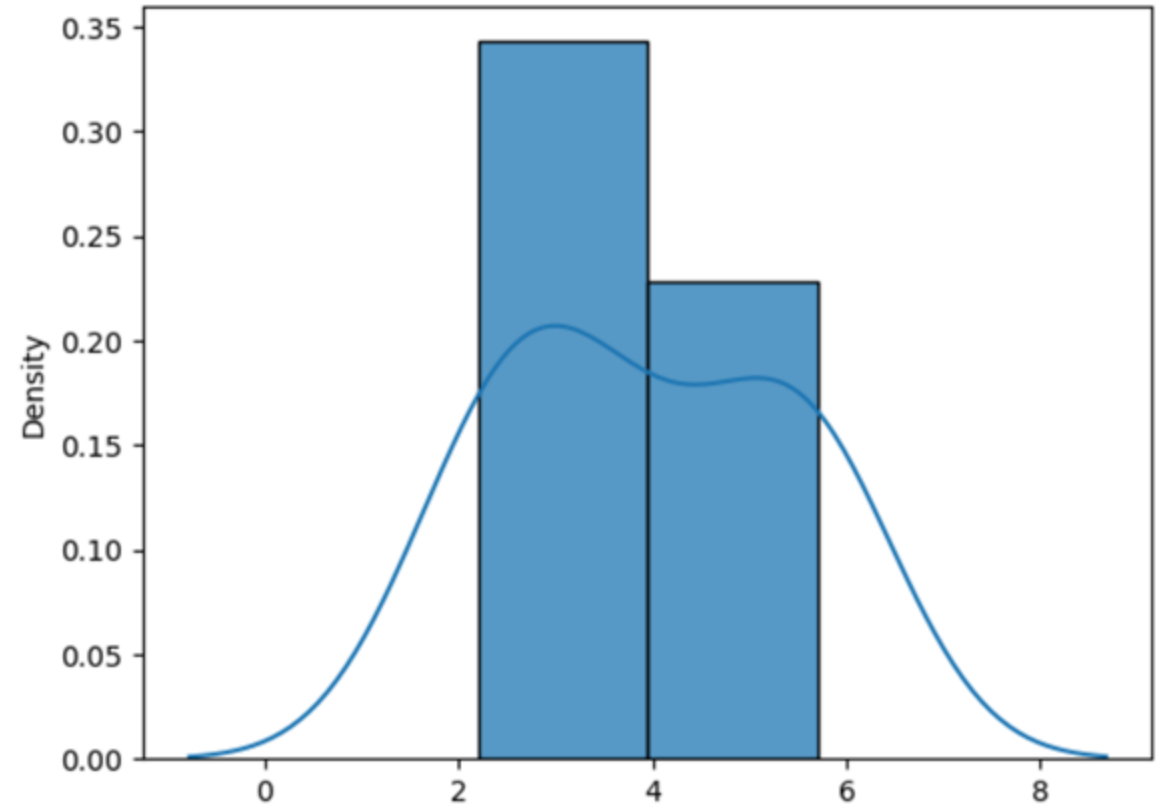
Result

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• A summary of the distribution



Each line represents a datapoint in the dataset (e.g. one country's HIV rate).

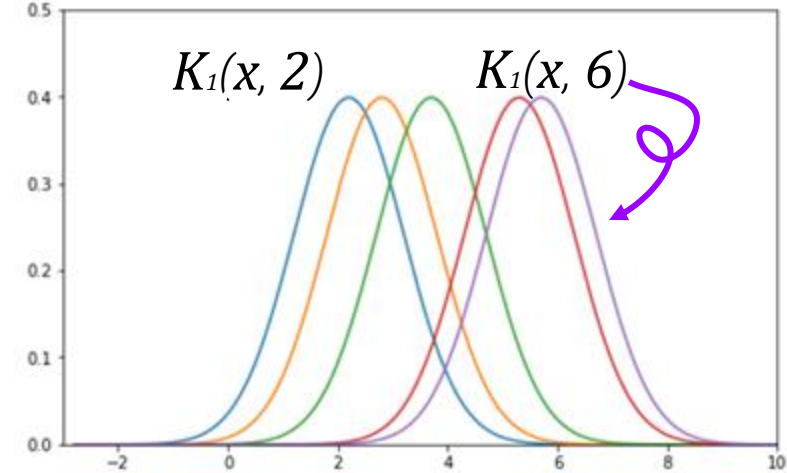


The density at each point corresponds to the KDE calculated based on kernels placed on all data points

Summary of KDE

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$$f_{\alpha}(x) = \frac{1}{n} \sum_{i=1}^n K_{\alpha}(x, x_i)$$



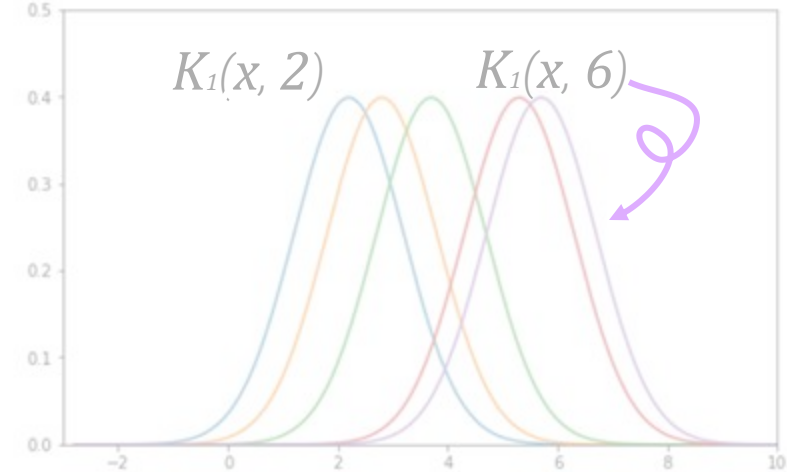
A general “KDE formula” function is given above.

- 1 $K_{\alpha}(x, x_i)$ is the function centered on the observation i .
 - Each kernel individually has area 1.
 - K represents our kernel function of choice. We’ll talk about the math of these functions soon.

Summary of KDE

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$$f_{\alpha}(x) = \frac{1}{n} \sum_{i=1}^n K_{\alpha}(x, x_i)$$



A general “KDE formula” function is given above.

- 1 $K_{\alpha}(x, x_i)$ is the **kernel** centered on the observation i .
 - Each kernel individually has area 1.
- 2 x represents any number on the number line. It is the input to our function.
- 3 n is the number of observed data points that we have.
 - We multiply by $1/n$ to normalize the kernels so that the total area of the KDE is still 1.

Kernels

A ~~kernel~~ (for our purposes) is a valid density function, meaning:

- It must be non-negative for all inputs.



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The most common kernel is the Gaussian = Normal distribution = bell curve = 1).

- Gaussian = Normal distribution = bell curve.
- Here, x represents any input, and x_i represents the i th observed value (datapoint).
- Each kernel is **centered** on our observed values (and so its distribution mean is x_i).
- α is the **bandwidth parameter**. It controls the smoothness of our KDE. Here, it is also the standard deviation of the Gaussian.

$$K_{\alpha}(x, x_i) = \frac{1}{\sqrt{2\pi\alpha^2}} e^{-\frac{(x-x_i)^2}{2\alpha^2}}$$

Memorizing this formula is less important than knowing the shape and how the bandwidth parameter α smoothes the KDE.

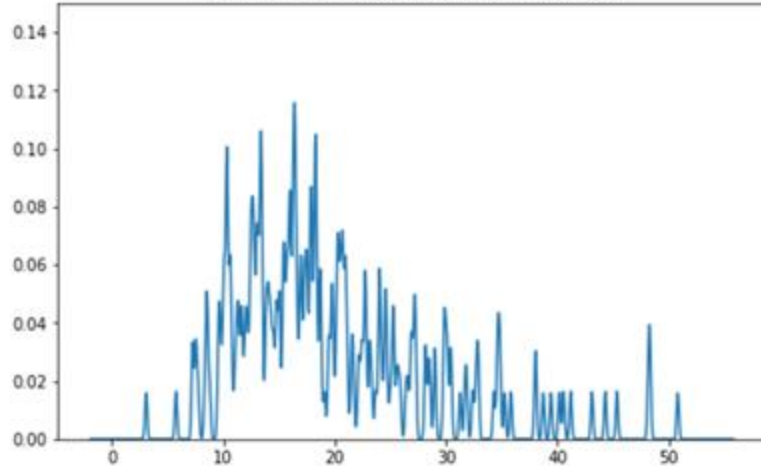
Effect of α on KDEs

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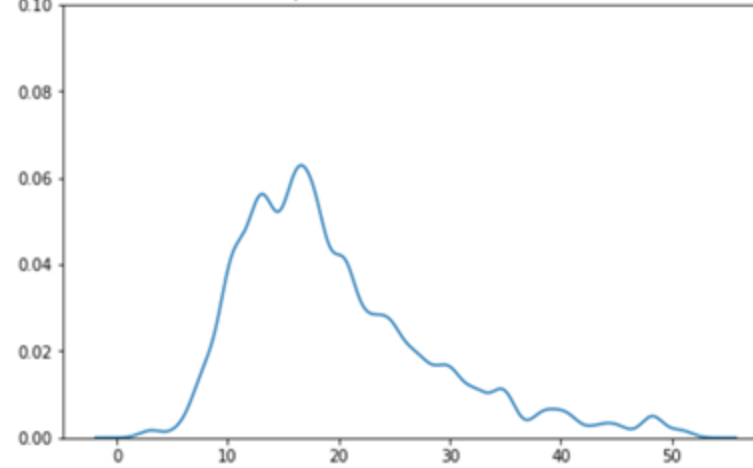
Bandwidth is analogous to the width of each bin in a histogram.

- As α increases, the KDE becomes more smooth.
- Large α KDE is simpler to understand, but gets rid of potentially important distributional information (e.g. multimodality).

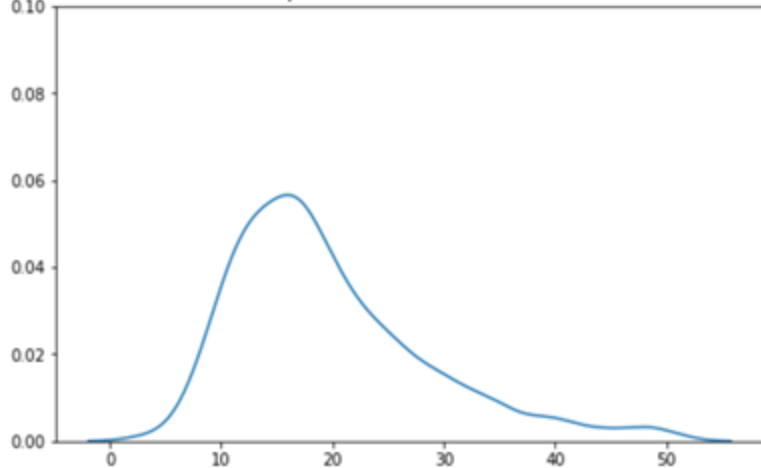
KDE of tips with Gaussian kernel and $\alpha = 0.1$



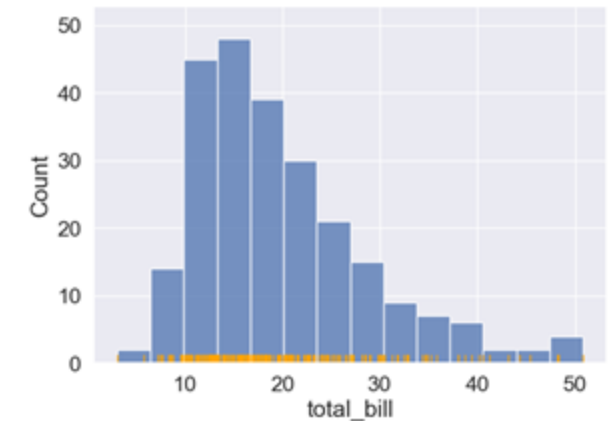
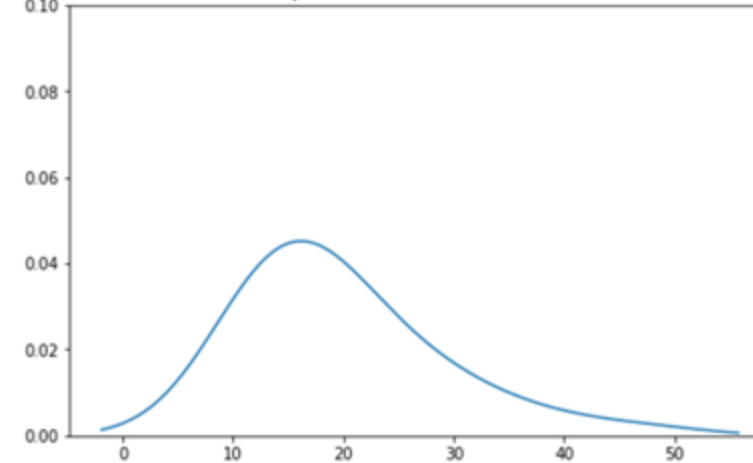
KDE of tips with Gaussian kernel and $\alpha = 1$



KDE of tips with Gaussian kernel and $\alpha = 2$



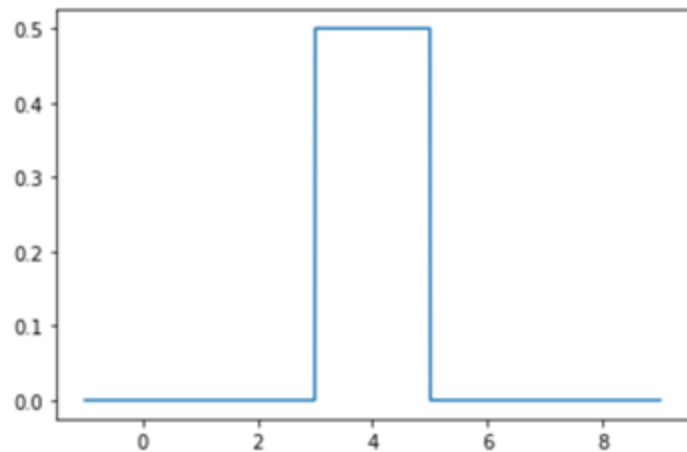
KDE of tips with Gaussian kernel and $\alpha = 10$



Other Kernels: Boxcar

As an example of another kernel, consider the **boxcar kernel**.

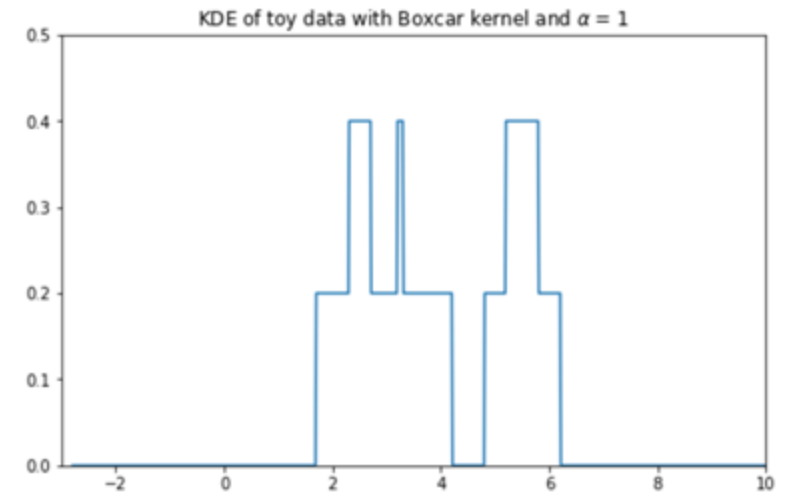
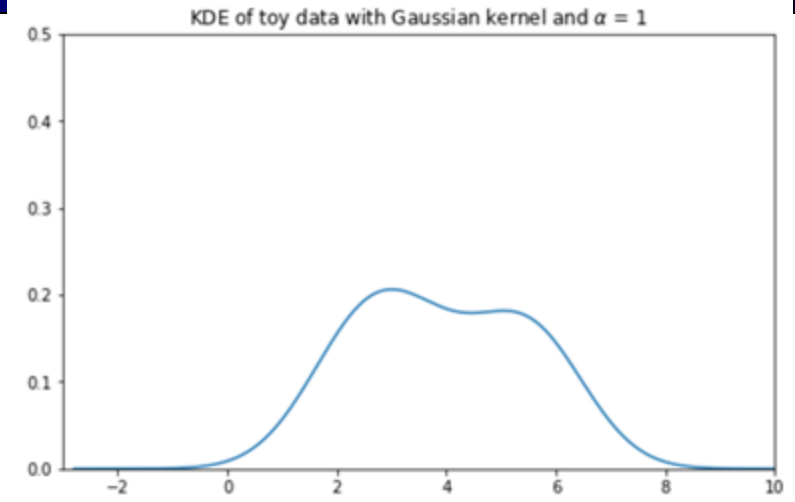
- It assigns uniform density to points within ϵ and 0 elsewhere.
$$K_{\alpha}(x, x_i) = \begin{cases} \frac{1}{\alpha}, & |x - x_i| \leq \frac{\alpha}{2} \\ 0, & \text{else} \end{cases}$$
- Resembles a histogram... *sort of*.



A boxcar kernel centered
on $x_i = 4$ with $\alpha = 2$.

- No Pre alternative.

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Which of the following are valid kernel density plots?

① Click **Present with Slido** or install our [Chrome extension](#) to activate this poll while presenting.

Have a Normal Day!

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